

The resolution of linguistic dating in Biblical Hebrew: calibrated ranges for the Pentateuchal sources, with cross-linguistic replication in Ancient Greek

Aaron Adair^{1*}

¹ Department of Physics, Massachusetts Institute of Technology, Cambridge, MA, USA

* Corresponding author: adairaar@mit.edu

Abstract

The linguistic dating of ancient texts from internal morphosyntactic evidence is central to biblical studies and classical philology, but its resolution — how finely a text can actually be placed in time — has never been measured. Reported accuracies vary by two orders of magnitude, in part because validation designs in this area frequently supply a text’s expected date to the model that is then asked to recover it. The present paper measures the resolution directly. Every out-of-sample result uses an agnostic prior; feature screening, standardisation and model fitting are performed inside each cross-validation fold; and significance is established by permutation of the date labels through the entire pipeline rather than by parametric assumption. Two model families are fitted and both are reported throughout. Applied to 25 units of the BHSA database spanning 760–167 BCE, the analysis yields a sharply bounded positive result. Hebrew morphosyntax carries a genuine diachronic signal: the models order text pairs chronologically with 69.8% and 67.8% accuracy across 295 pairs, against a permutation null of 47.6% and 43.6% ($p = 0.022, 0.029$), and leave-one-out Spearman correlations between true and predicted dates are +0.58 and +0.49 ($p = 0.016, 0.024$). That signal does not, however, support absolute dating: leave-one-out mean absolute error is 121–156 yr against 137 yr for the trivial predictor that assigns every text the corpus mean date, and four-way period assignment reaches only 40–48%. We show that previously reported holdout accuracies of order 10–30 yr in this literature, including in an earlier version of this work, arise because each held-out text is dated under a prior centred on its own scholarly date; for the most tightly anchored texts the linguistic data contributes under 6% of posterior precision. Applied to the Pentateuchal sources with distribution-free conformal intervals, the framework supports one substantive conclusion — the Priestly, Deuteronomic and JE composites are post-exilic with probability ≥ 0.92 under both families — while declining to place them within a particular post-exilic period. The Song of the Sea and Song of Deborah emerge as the two earliest of nineteen undated units in both families without having been trained on. Code and corpora are publicly available (see Data Availability).

Introduction

Scholars in biblical studies, classical philology, and Near Eastern linguistics routinely use internal linguistic evidence to constrain the dates of ancient texts. The transition from Classical Biblical Hebrew (CBH; also called Standard Biblical Hebrew, SBH) to Late Biblical Hebrew (LBH) is among the best-documented diachronic shifts in any ancient language [1–3], and comparable transitions are well attested in Ancient Greek [4] and Akkadian. Despite this rich empirical tradition, linguistic dating almost never produces calibrated probability estimates; scholars identify features as “archaic” or “late” and assign texts to qualitative categories, but the resulting claims carry no quantified uncertainty.

A recent contribution by Faigenbaum-Golovin et al. [5] demonstrated that statistical analysis of lemma n-gram frequencies can reliably distinguish among the three major scribal corpora of the Hebrew Bible — the old layers of Deuteronomy (D), the Deuteronomistic History (DtrH), and the Priestly source (P) — achieving approximately 84% chapter-level attribution accuracy on a 50-chapter benchmark, with interpretable discriminating-lemma lists linking computational results to traditional philological categories. That study establishes the viability of quantitative computational methods for biblical Hebrew scholarship and demonstrates that linguistically distinct scribal schools leave detectable statistical fingerprints even in relatively short texts. Its focus, however, is *authorship attribution*: given a passage of disputed origin, which known scribal milieu produced it? The present paper addresses the complementary but methodologically distinct question of *diachronic dating*: not “which school wrote it?” but “when was it written?” These are related but separable questions that demand different statistical frameworks, different feature sets, and different validation strategies, as detailed below.

This paper addresses two methodological gaps that authorship-attribution approaches leave open. First, no existing probabilistic framework provides date posteriors with credible intervals for Biblical Hebrew texts. Qualitative assessments of individual features are genuinely informative, but they cannot be combined into a single coherent posterior without a model that propagates feature-level evidence into a joint probability distribution over dates. Second, no principled framework separates the temporal signal from two well-known confounds: *archaism* (a late composer deliberately employing archaic vocabulary and morphology) and *genre* (legal or narrative register producing feature distributions that mimic either earlier or later stages of the language independently of composition date).

The present framework contributes the following:

1. A morphosyntactic feature extraction pipeline operating on the BHSA/ETCBC database [6, 7], producing 35 features across three tiers (lexical, morphological, and clause-level syntactic), selected by Spearman correlation ($p < 0.30$) with 100% LOO sign-consistency.
2. The MLE-MVN Bayesian dating model: per-feature OLS regression, Tikhonov-regularized residual covariance, and posterior over a 500-point date grid with explicit uncertainty quantification.
3. The HB-VI model: hierarchical Bayesian regression with mean-field variational inference (MFVI) and ADAM optimization [8, 9], providing proper parameter uncertainty propagation and soft register assignment.
4. Prior sensitivity analysis distinguishing data-driven from prior-dominated date estimates, and a decomposition of posterior precision that quantifies how much of a reported date is contributed by the linguistic evidence rather than by the prior.
5. A leave-one-out protocol in which feature screening, standardisation and model fitting are performed inside each fold, and every out-of-sample unit is dated under an agnostic prior.
6. Permutation inference: the date labels are shuffled and the entire pipeline re-run, so the null distribution absorbs selection effects.
7. Distribution-free conformal prediction intervals calibrated on leave-one-out residuals, replacing parametric intervals whose empirical coverage is roughly three-quarters of nominal.

The results are bounded in both directions. Hebrew morphosyntax carries a genuine diachronic signal: under leakage-free cross-validation the models order text pairs chronologically with 69.8% and 67.8% accuracy across 295 pairs against a permutation null near 45% ($p = 0.022, 0.029$), and rank correlations between true and predicted date reach +0.58 and +0.49 in two independently specified model families. The same analysis shows that this signal does not support absolute dating: leave-one-out error of 121–156 yr must be read against 137 yr for the trivial predictor assigning every text the corpus mean date, and four-way period assignment reaches 40–48%. We further show that holdout accuracies of order 10–30 yr previously reported for these models — including by us — arise because each held-out text is dated under a prior centred on its own scholarly date; for the most securely anchored texts the linguistic evidence contributes under 6% of posterior precision. Applied to the Pentateuchal sources under corrected conditions, the framework establishes that the Priestly, Deuteronomic and JE composites are post-exilic with probability ≥ 0.92 under both model families, while declining to locate them within a particular post-exilic period.

The paper is organized as follows. The Background section reviews the CBH/LBH distinction and the methodological debates surrounding linguistic dating. The Data section describes the training corpus and feature extraction. The MLE-MVN and HB-VI sections present the two Bayesian models. The Prior Sensitivity Analysis section covers the three-mode sensitivity framework and the model-independent feature scoring used to make the inputs auditable. The Validation section documents the prior-leakage failure mode, sets out the leave-one-out protocol, and reports what the signal does and does not support. The Results section applies the corrected framework to the undated units. The Discussion section addresses limitations, extensibility, and the relationship to authorship-attribution approaches.

Background: diachronic Biblical Hebrew

The CBH/LBH distinction

Classical Biblical Hebrew (CBH) is the language of the pre-exilic prophetic corpus and early narrative material (8th–7th centuries BCE): Amos, Hosea, early Isaiah, and the narrative strands of Samuel-Kings. Late Biblical Hebrew (LBH) characterizes the post-exilic prose of Ezra, Nehemiah, Chronicles, Esther, and Daniel (5th–2nd centuries BCE).

The transition is marked by well-studied lexical, morphological, and syntactic changes [2, 3, 10–12]. At the lexical level, the first-person singular pronoun shifts from *ankī* (CBH) to *anī* (LBH); the relative pronoun *ašer* gives way to the enclitic *šē-*; the negator *lō'* is supplemented by and eventually partly replaced by *'ēn*. At the morphological level, the wayyiqtol (narrative imperfect) declines in frequency while the *hāyāh* + participle periphrastic progressive increases. At the syntactic level, infinitive-absolute constructions rise, and the overall sentence structure shifts toward the analytic patterns associated with Aramaic influence [13, 14].

The consensus view [1, 15–17] holds that these features shift monotonically along the temporal axis, constituting a reliable if noisy diachronic signal. The present study adopts this view as a working assumption while formally modeling and partially testing the alternatives.

The diachronic versus dialectal debate

Young, Rezetko, and Ehrensverd [13] argue that the CBH/LBH contrast may reflect geographic dialect, scribal tradition, or literary register rather than chronology. The Qumran sectarian texts (Dead Sea Scrolls, ~150 BCE) deliberately employ CBH vocabulary and morphology while simultaneously using Aramaic loan-words — a pattern interpreted as literary register rather than temporal reality.

Hurvitz and the diachronic mainstream [10] counter that the dialectal alternatives require additional auxiliary hypotheses; and that the Qumran archaism case presupposes the original CBH features had already receded, confirming rather than undermining the diachronic interpretation.

For the present paper, the question is not which side of this debate is correct but whether the empirical evidence can distinguish between them. The framework operationalizes the distinction rather than adjudicating it on a priori grounds. One natural operationalization is to ask whether clause-level syntactic patterns, which fall below the level of deliberate stylistic choice, agree with a date derived from lexical and morphological features that a competent imitator could manipulate; systematic disagreement would be a

falsifiable indicator of archaism. We implemented this test and report under Results: dating the undated units that it did not function on the present corpus, flagging no features on any unit. The archaism question therefore remains open here, and the distinction between an early text and a skilled archaizing one is one this analysis cannot draw.

The circularity problem

Dating methods that use features identified by comparing texts of assumed date risk circularity: the assumed dates shape the feature set, and the feature set is then used to infer unknown dates. Three safeguards are adopted.

First, the training corpus is restricted to prophetic books with *independent* historical anchoring via synchronisms with datable rulers in Assyrian, Babylonian, and Persian king-lists. Torah texts are excluded from training entirely, so all Torah date estimates are genuine out-of-sample predictions.

Second, a held-out validation set (oracle Jeremiah chapters) was withheld from all feature selection and training. Its correct recovery by the model (see the Validation section) provides empirical evidence against the circularity objection.

Third, the prior sensitivity analysis (see the Prior Sensitivity Analysis section) explicitly quantifies how much of each HB-VI date is determined by the scholar-specified prior versus the linguistic data.

Beyond these three safeguards, the choice of the prophetic corpus as the training base deserves explicit justification, and the *direction* of any residual misdating error is methodologically important.

Why the prophetic corpus. The Hebrew prophetic books offer a form of historical anchoring available nowhere else in the Hebrew Bible: their superscriptions and internal chronological notices can be cross-checked against records entirely external to the biblical tradition. Ezekiel explicitly dates thirteen of his visions to regnal years of the Babylonian exile — “the fifth year of King Jehoiachin’s captivity” (Ezek 1:2), “the sixth year” (8:1), and so forth — which can be correlated with Babylonian administrative texts that independently date Jehoiachin’s deportation to 597 BCE. Other eighth- and seventh-century prophets (Amos, Hosea, Isaiah, Micah, Zephaniah, Nahum) are anchored by superscriptions naming Judahite and Israelite kings whose reigns are independently documented in the Assyrian and Neo-Babylonian annals. No comparable external anchoring exists for the Torah or the poetic books: their dates are derived entirely from literary-critical arguments, which is precisely the kind of circularity the present study seeks to avoid. The prophetic corpus therefore represents the most defensible choice for a training set whose dates are genuinely independent of the linguistic features under analysis.

Asymmetric error and conservative upper limits. If the accepted dates for the prophetic training texts were wrong, the error would be asymmetric: these texts can be *no older* than the historical events they reference (the superscriptions and Ezekielian dates provide firm *termini post quos*), but they could in principle be *younger* — composed or reaching final form some time after the events they describe. This asymmetry has a direct consequence for the date estimates produced by the model. A training corpus that is misdated in the direction of being too early would calibrate the feature-to-date mapping at too early a point on the temporal axis; unknown texts dated by reference to that calibration would also be estimated as too early, meaning their true dates would be *later* than the model reports. In other words, if the prophetic training dates carry any systematic error, that error propagates in a single direction: all model-derived dates shift toward greater recency, not greater antiquity. The model’s output for a text such as the Song of the Sea — whose predictive mass under both model families is the earliest of any undated unit — would be pushed later still if the training corpus were found to be younger than traditionally assumed. The dates reported in this study should therefore be understood as *conservative upper bounds on antiquity*: a text dated here to 400 BCE may be more recent, but the linguistic evidence as calibrated against the prophetic corpus provides no support for a date earlier than that estimate.

Prior computational work: authorship attribution versus diachronic dating

Computational approaches to textual attribution and stylometric dating have a substantial prior history, from Mosteller and Wallace’s Bayesian analysis of the Federalist Papers [18] through the stylometric tradition surveyed by Holmes [19] and the function-word methods of Burrows [20]. Among work applied

specifically to the Hebrew Bible, Faigenbaum-Golovin et al. [5] provide the most methodologically rigorous prior computational treatment. Their Higher Criticism (HC) discrepancy method [5] tests whether the distribution of lemma n-gram frequencies in a target text deviates from the null hypothesis that it was produced by the same process as a reference corpus. Applied to 50 ground-truth chapters spanning D, DtrH, and P, the method achieves 84% attribution accuracy in leave-one-out validation and yields interpretable discriminating-lemma lists that directly engage traditional philological categories. The statistical optimality properties of the HC statistic (sensitivity to rare but consistent frequency deviations across a large vocabulary) make it well-suited to its task.

The present framework is designed to answer a different question. Five methodological differences follow from that distinction.

Continuous date posteriors versus categorical corpus assignment. The HC method assigns each text to the most likely of three corpora. Because D and DtrH are both dated to the late monarchic or exilic period, two texts composed at 650 and 590 BCE receive identical labels if both score closest to DtrH; no intra-school temporal resolution is possible. The MLE-MVN and HB-VI models instead compute a posterior probability distribution over a 500-point continuous date grid, recovering whatever temporal resolution the data support and quantifying uncertainty explicitly via credible intervals.

Morphosyntactic features versus raw lemma frequencies. Faigenbaum-Golovin et al. operate on raw lemma and lemma n-gram frequencies — a bag-of-words representation in which content words (e.g., *gold* in Tabernacle chapters, *king* in regnal narratives) dominate the signal. Content-word frequencies are sensitive to *topic* and *genre* as much as to *authorship*, and they carry little temporal signal for cross-genre comparisons. The 35 features used in this study are drawn from a century of diachronic Hebrew linguistics scholarship [1–3, 13] precisely because they are sensitive to *period* rather than *topic*: the *ankī*→*anī* pronoun shift, the negator transition, the wayyiqtol decline, and the infinitive-absolute rate change systematically across time and resist topical or generic influence. This distinction is especially important for cross-genre comparisons: bag-of-words features cannot compare poetic or legal texts against narrative texts without large genre confounds.

Archaism diagnostic. The HC method provides no mechanism for distinguishing a genuinely ancient text from one composed in a deliberately archaic register. A Persian-period author who archaizes vocabulary to D/DtrH norms will be assigned to DtrH regardless of actual composition date. The resistant/full-model divergence introduced in Section 7 detects this pattern directly: surface features under conscious compositional control (lexicon, morphology) and deep clause-level syntactic patterns that operate below deliberate stylistic choice are evaluated independently, and systematic divergence between their date estimates constitutes a quantified, falsifiable indicator of archaism. This diagnostic has no counterpart in the HC framework.

External holdout validation. Faigenbaum-Golovin et al. validate via leave-one-out and k -fold cross-validation within the three training corpora. This confirms internal consistency but cannot test whether the method recovers *correct* calendar dates, because the ground-truth labels are scribal-school assignments rather than independently anchored absolute dates. The present framework is validated against texts anchored to external historical events — oracle Jeremiah chapters to the Neo-Babylonian period [21], Haggai to 520 BCE (Persian regnal date), Daniel Hebrew chapters to 167 BCE (Maccabean crisis) — providing a genuinely independent test that the temporal signal is real rather than self-referential.

Cross-linguistic generalization. A 63-text Ancient Greek corpus spanning 440 BCE–360 CE was re-analysed under the identical protocol (Fig 1B,C). The originally reported holdout recoveries “within ~30 yr” were produced by the same prior-leakage design documented under Validation and are withdrawn. What replaces them is weaker but genuinely cross-linguistic: pairwise ordering accuracy of 60.3% and 58.9% against a permutation null near 40% ($p = 0.037, 0.050$), with leave-one-out rank correlations of +0.36 and +0.27. The ordinal signal replicates in an unrelated language at comparable significance, and that is the generalization claim the data support.

The negative result replicates more sharply still. Greek leave-one-out MAE is 294.5 yr (generative) and 205.4 yr (ridge) against 210.9 yr for the constant predictor: one family is substantially worse than a constant and the other improves on it by five years ($p = 0.24$). Corrected holdout errors are 339.0 and 171.2 yr, against the 8.7 yr originally reported. That this occurs at $n = 63$ over an 800-year span — where a small corpus cannot be blamed — indicates the limitation is intrinsic to the approach rather than a consequence of

sample size.

These five differences are differences of purpose rather than deficiencies in either approach. Authorship attribution and diachronic dating ask related but separable questions; the ideal analysis of any disputed text would apply both. The HC output (scribal-school membership) can in principle be used to set register priors in HB-VI, and conversely HB-VI date posteriors constrain the plausible compositional settings for HC attribution decisions. This complementarity is taken up in the Discussion.

Data: corpus and feature extraction

The BHSA database

All Hebrew text analysis uses the Biblia Hebraica Stuttgartensia Amstelodamensis (BHSA) database, version 2021, produced by the Eep Talstra Centre for Bible and Computer (ETCBC) at Vrije Universiteit Amsterdam [6, 7, 22]. BHSA provides morphological tagging at the word level, phrase-level function annotation, clause boundary segmentation, and lexeme coding for the entire Hebrew Bible. Data are accessed programmatically via the Text-Fabric Python library, enabling reproducible extraction without manual annotation. No Aramaic chapters are included in training or testing.

Training corpus

The dated corpus comprises 25 units spanning 760–167 BCE and 146,059 tokens: 23 training units and two units held out of both models (Table 1). Date anchoring uses explicit synchronisms with datable ancient Near Eastern events wherever these are available. Amos 1:1 places that book in the reigns of Uzziah and Jeroboam II (~760 BCE); Zechariah 1–8 and Haggai carry explicit Darius regnal dates (518 and 520 BCE); Ezekiel supplies thirteen regnal dates checkable against Babylonian records; Daniel’s Hebrew chapters (1, 8–12) are anchored to 167 BCE by the Maccabean crisis. Daniel is restricted to those Hebrew chapters; the Aramaic chapters (2–7) are a different language and are excluded.

Not every unit is anchored this securely, and the corpus does not pretend otherwise. Where a date rests on literary-critical rather than synchronistic grounds, σ is widened accordingly: Joel and Zechariah 9–14 carry $\sigma = 150$ yr, Ecclesiastes 120 yr, Isaiah 56–66 100 yr, and Obadiah 80 yr, against 10–30 yr for the synchronistically anchored texts. This is a substantial widening relative to earlier versions of this corpus, and it is deliberate: a narrow prior on a literary-critical date imports the very assumption the analysis is meant to test. The prior sensitivity analysis below quantifies what that assumption is worth.

Two units are held out of both models and excluded from all feature screening: the Jeremiah oracular stratum (chs. 1–6, 8–10, 12–16, 19–20, 22–23, 30–31, 46–51) and Haggai. Whole-book Jeremiah is excluded from training entirely, because it contains the oracular stratum and would otherwise leak the holdout into the training set. The Deuteronomistic prose stratum of Jeremiah (chs. 7, 11, 17–18, 21, 24–29, 32–45, 52) is treated as an undated target and never trained on. Habakkuk and Daniel are retained in MLE-MVN training but held out of HB-VI.

A note on what “held out” must mean. In the analyses reported below, a held-out unit contributes nothing to feature screening, to standardisation, to regression fitting, or to its own prior. The last of these is not a formality. A model that receives a text’s scholarly date as a prior mean will return approximately that date whatever the text contains, and will do so most emphatically for the texts whose dates are best established — that is, precisely the texts on which a validation exercise appears most convincing. Section Validation quantifies this effect for the present corpus. All out-of-sample results in this paper use an agnostic prior.

S5 Table lists word counts for all units, including the Documentary Hypothesis source composites, which are excluded from both training and validation and are dated solely from the model.

Feature extraction

Features are extracted at three tiers that differ in their susceptibility to deliberate stylistic manipulation (Table 3).

Table 1. Dated corpus: 23 training units and 2 held-out units. Date (BCE) is the scholarly consensus midpoint used as the training label; σ is the accompanying uncertainty. Sigmas were widened by approximately a factor of two, relative to earlier versions of this corpus, on every text whose date rests on literary-critical rather than synchronistic grounds. Register: S = SBH (Standard Biblical Hebrew, equivalent to CBH), T = Transitional, L = LBH. Word counts are token counts from the BHSA extraction used for all feature rates. ^aHeld out of the HB-VI model only. ^bHeld out of both models, and excluded from all feature screening. Whole-book Jeremiah is excluded from training entirely, because it contains the Jeremiah oracle holdout.

Unit	Register	Date (BCE)	σ (yr)	Words
<i>Training</i>				
Amos	S	760	20	2,780
Hosea	S	745	25	3,146
Micah	S	725	25	1,895
Isaiah 1–39	S	720	30	13,504
Nahum	S	650	25	746
Zephaniah	S	630	20	1,037
Habakkuk ^a	S	605	20	897
Obadiah	T	585	80	392
Ezekiel	T	580	20	26,182
Lamentations	T	570	50	1,945
Isaiah 40–55	T	545	20	5,738
Zechariah 1–8	T	518	10	2,484
Malachi	T	460	25	1,187
Joel	T	400	150	1,318
Jonah	T	400	60	985
Isaiah 56–66	T	400	100	3,689
Ezra	L	380	60	5,268
Nehemiah	L	380	60	7,842
Zechariah 9–14	L	350	150	1,987
Chronicles	L	330	40	35,330
Esther	L	300	60	4,621
Ecclesiastes	L	250	120	4,233
Daniel (Heb.) ^a	L	167	10	3,437
<i>Held out of both models</i>				
Jeremiah oracle ^b	S	605	30	14,539
Haggai ^b	T	520	10	877
Total				146,059

Tier 1 (lexical, 19 features). Rates per 1,000 words of specific function words, particles, demonstratives, and prepositions. Examples: the *ašer*/*še* ratio; the *anḵī*/*anī* ratio; the absolute rate of the particle *gam*; the rate of the accusative marker *et*. Tier 1 features most directly capture the CBH/LBH lexical axis but are also most susceptible to deliberate archaism, because a competent composer can deliberately choose archaic function words.

Tier 2 (morphological, 12 features). Verb stem fractions (qal, piel, hif, niphal as fractions of all verbs); verb form fractions (wayyiqtol, qatal, yiqtol, participle, imperative fractions); absolute noun ratio; pronominal suffix rate. These are harder to archaize wholesale but remain subject to word-level editorial manipulation.

Tier 3 (clause/phrase, 4 features). Four clause-level features, the least susceptible of the three tiers to deliberate manipulation: the infinitive-construct fraction (**frac_inf**), the fronted-clause fraction (**frac_fronted**), the null-subject fraction (**frac_null_subj**), and the waw-qatal fraction (**frac_wqt1_wayq**). These measure syntactic templating patterns that are acquired below the level of conscious stylistic choice and are therefore most resistant to deliberate manipulation.

Feature selection. Spearman rank correlation with date is computed for each candidate feature against the 22-text training set. Features are retained at a permissive significance threshold of $p < 0.30$: with only $N = 22$ training texts, a strict Benjamini-Hochberg FDR correction [23] at $\alpha = 0.05$ would retain only ~ 10 features, which is insufficient to capture the multi-dimensional CBH/LBH signal documented over a century of Hebrew linguistics scholarship. The $p < 0.30$ threshold is paired with a leave-one-out (LOO) robustness filter: a feature is retained only if its Spearman ρ preserves sign in $\geq 68\%$ of LOO resamplings (the 68% threshold corresponds to one standard deviation of a binomial sign-consistency test, providing a principled stopping criterion). In practice, all 35 features surviving the $p < 0.30$ screen achieve 100% LOO sign consistency (LOO fraction = 1.00), well above the 68% floor, confirming that the selected features carry a robust diachronic signal. This yields a final set of **35 features**. Table 2 lists the ten features with the largest $|\rho|$; the complete set is provided in S1 Table. OLS slopes ($\hat{\beta}_j$) and their confidence intervals for all 35 features are shown in S6 Fig.

Table 2. Top-10 features by Spearman $|\rho|$ with date. All have LOO fraction = 1.00. Sign of ρ indicates direction: negative = feature declines with date (more CBH-like in older texts). Full 35-feature list in S1 Table.

Rank	Feature	Tier	Spearman ρ	Direction
1	<i>ankī/ani</i> pronoun ratio	1	−0.94	Declines
2	Wayyiqtol rate	2	−0.91	Declines
3	<i>ašer</i> relative rate	1	−0.89	Declines
4	<i>še-</i> relative fraction	1	+0.87	Rises
5	<i>lo'/'en</i> negator frac.	1	+0.85	Rises
6	Qatal rate	2	+0.83	Rises
7	Inf.-construct fraction [R]	3	−0.81	Declines
8	Niphal fraction	2	+0.79	Rises
9	Fronted-clause fraction [R]	3	+0.77	Rises
10	<i>gam</i> rate	1	−0.75	Declines

[R] = Tier-3 resistant-model feature.

Table 3. Feature tier summary. J = number of features in tier. Susceptibility indicates the degree to which deliberate archaism can manipulate features within that tier.

Tier	Type	J	Archaism susceptibility
1	Lexical (function words, particles)	19	High
2	Morphological (verb forms, stems)	12	Medium
3	Clause/phrase syntax	4	Low
Total selected		35	

Word n-gram model

As an independent grammatical-sequence dating instrument, POS-tag n-gram features [24, 25] are extracted that are immune to topic variation and largely immune to deliberate lexical archaism: a composer can choose archaic vocabulary but cannot easily rewrite every grammatical-sequence pattern to match an earlier era.

Two feature types are used. **Type A** features are POS-tag bigrams and trigrams on all words (e.g., *verb-wayyiqtol · noun*, *prep · prps*), capturing deep grammatical flow. **Type B** features are function-word lexeme bigrams and trigrams restricted to seven closed-class POS categories (prepositions, conjunctions, articles, negators, personal pronouns, demonstratives, interrogatives), making them robust to both topic and rare-vocabulary variation. The combined A+B model (MAP_{AB}) uses 56 features and is the primary word n-gram estimate reported in all tables.

Character n-gram model: assessment and exclusion. An earlier version of the framework included character 3-gram and 4-gram frequencies. Reliability testing on 1QS (Community Rule, Dead Sea Scrolls; independently dated ~ 150 BCE) returned MAP = 760 BCE — a 600-year error attributable to deliberate

orthographic archaism. Sub-morphemic patterns are not immune to deliberate archaism in the Biblical Hebrew corpus. The character n-gram model is therefore excluded from all analyses; this negative result is reported because it is informative about the limits of the approach.

The MLE-MVN model

Per-feature OLS regression

Each feature f_j ($j = 1, \dots, J$, $J = 35$) is modeled as a linear function of date d plus Gaussian noise:

$$f_j(d) = \alpha_j + \beta_j d + \varepsilon_j, \quad \varepsilon_j \sim \mathcal{N}(0, \sigma_j^2). \quad (1)$$

Parameters $(\hat{\alpha}_j, \hat{\beta}_j)$ are estimated by OLS on the 22-unit training corpus. The predicted feature vector at date d is $\boldsymbol{\mu}(d) = \hat{\alpha}_j + \hat{\beta}_j d$ stacked over j .

The linear model is a simplification: some features (e.g., wayyiqtol fraction) may have mild non-linear trends. A sensitivity analysis with quadratic terms shows negligible improvement over the date range 760–167 BCE, justifying the linear approximation.

Residual covariance and regularization

Earlier versions of this work motivated Tikhonov regularization of the residual covariance by rank deficiency: with J features and N training texts, $J > N$ makes $\hat{\Sigma}$ singular and a ridge term $\lambda \mathbf{I}$ necessary for invertibility. That justification no longer applies and is replaced here.

The power analysis of the present corpus (S2 Table) shows that a screening threshold retaining 35 features at $N = 23$ carries an estimated false discovery proportion above 0.5. The defensible operating point is $\alpha = 0.05$, retaining 14 features, with Benjamini-Hochberg at $q = 0.10$ retaining 7 as a conservative band. At $J = 14 < N = 23$ the sample residual covariance is no longer rank-deficient, and regularization is not required for invertibility.

Regularization is nonetheless retained, for a different and better reason. The residual covariance is estimated from 23 observations and is correspondingly noisy; its smallest eigenvalues are estimated worst, and the likelihood of Eq. 2 weights each direction by the inverse of its eigenvalue. Shrinking the spectrum toward its mean therefore controls the influence of directions the data cannot resolve. We select λ by inner leave-one-out on the training folds rather than fixing it a priori. The multiverse analysis reported under Validation shows why this matters: without shrinkage, adding chance-correlated features narrows credible intervals rather than widening them.

Posterior computation

A 500-point date grid spanning 900 BCE to 100 CE (step ≈ 2 yr) is evaluated. The log-likelihood of observed feature vector $\mathbf{x}$ at grid point d under a multivariate Gaussian model [26] is

$$\ell(d) = -\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}(d))^{\top} \boldsymbol{\Sigma}_{\text{reg}}^{-1}(\mathbf{x} - \boldsymbol{\mu}(d)). \quad (2)$$

A weakly informative Gaussian prior $p(d) = \mathcal{N}(464, 300^2)$ (centered at the training midpoint) is added in log space, and the result is normalized to a proper posterior:

$$p(d \mid \mathbf{x}) \propto \exp\{\ell(d)\} \cdot p(d). \quad (3)$$

The MAP estimate is the mode; 68% and 95% highest-density credible intervals (HDCIs) are extracted numerically [27].

The HB-VI model

Motivation and limitations of MLE-MVN

The MLE-MVN model treats each feature regression independently and uses point estimates for all parameters. Two failure modes motivate the hierarchical extension. First, the resulting 68% CIs are extremely narrow (median 3–4 yr for SBH texts), indicating the model is systematically overconfident; parameter uncertainty in $\hat{\beta}_j$ is not propagated. Second, register assignment is hard: a text must be pre-assigned to one of three groups (SBH/Transitional/LBH) before the posterior is computed. Misassignment cascades into catastrophically wrong dates (e.g., Haggai assigned to LBH: MAP = 361 BCE vs. 520 BCE actual; error = 159 yr; holdout MAE for MLE-MVN = 134 yr).

Hierarchical model architecture

The HB-VI model is a hierarchical Bayesian regression in which per-group feature loadings are drawn from a global distribution:

$$\beta_{r,j} \sim \mathcal{N}(\mu_{\beta_j}, \sigma_{\beta}^2), \quad (4)$$

where $r \in \{1, 2, 3\}$ indexes the register group (SBH, Transitional, LBH), j indexes the feature, and μ_{β_j} and σ_{β} are global hyperparameters shared across groups. Groups share statistical strength via the global hyperparameters while maintaining group-specific slopes. Intercepts $\alpha_{r,j}$, noise variance $\sigma_{r,j}$, and the global hyperparameters are all variational parameters.

The training corpus for the HB-VI model uses 19 texts (7 SBH, 6 Transitional, 6 LBH), with the three holdout texts (Habakkuk, Haggai, Daniel) excluded, and the full Jeremiah book replaced by the oracle Jeremiah chapters as the SBH anchor. Date priors for training texts are $\mathcal{N}(d_{\text{BCE}}, \sigma_{\text{date}}^2)$, directly encoding known date uncertainty.

Variational inference training

Full posterior sampling is computationally prohibitive at this model size, so mean-field variational inference (MFVI) is used instead [28, 29]: the posterior is approximated by an independent Gaussian over each parameter, with means and variances optimized to minimize KL divergence from the true posterior. This converts inference into an optimization problem tractable on standard hardware without GPU dependency.

Formally, the posterior $p(\boldsymbol{\theta} \mid \mathbf{X})$ is approximated by MFVI: the variational family $q(\boldsymbol{\theta}) = \prod_i q(\theta_i)$ factorizes over all parameters. The Evidence Lower Bound (ELBO) is maximized:

$$\mathcal{L}(q) = \mathbb{E}_q[\log p(\mathbf{X} \mid \boldsymbol{\theta})] - \text{KL}(q(\boldsymbol{\theta}) \parallel p(\boldsymbol{\theta})). \quad (5)$$

Optimization uses ADAM [8] (learning rate = 0.02, 3,000 iterations) with the reparameterization trick for gradient flow through Gaussian sampling. Convergence is declared when the ELBO improvement over 100 iterations falls below 0.01%; in practice convergence occurs within $\sim 2,000$ iterations. The implementation uses the `autograd` library [30] to avoid GPU dependency.

Date posteriors and best-group selection

For a text with known register, the date posterior is computed by evaluating the group-specific log-likelihood over a 1,000-point BCE grid, integrating over parameter uncertainty via Monte Carlo ($N_{\text{MC}} = 200$ samples from the variational posterior).

For texts with unknown register (documentary sources D, P, JE; short sub-sources), *best-group selection* is used: the marginal log-likelihood is computed for all three groups via `logsumexp` over the date grid, and the group maximizing marginal log-likelihood is selected. This avoids the hard pre-assignment failure mode of MLE-MVN.

Training ceiling effect. The HB-VI model, like MLE-MVN, cannot reliably extrapolate beyond the training range (~ 760 – 330 BCE). Archaic texts whose features cluster with early SBH are mapped to the training ceiling (~ 760 BCE), and the model can only report “consistent with the oldest SBH anchor,” not a pre-monarchic date. This limitation is acknowledged in all results involving archaic poetry.

Pentateuchal source dating

Source-composite results from both model families are reported together under Results: dating the undated units, after the validation analysis that determines how they should be read. We defer them deliberately. The empirical results of this section in earlier versions of this work were computed under a design in which each held-out unit was dated using its own scholarly date as a prior; the resulting figures are not comparable to those reported here and are not repeated.

Prior sensitivity analysis

The prior sensitivity analysis that follows was originally conceived as a robustness check. In light of the leakage analysis under Validation it carries more weight than that: the degree to which a posterior is prior-dominated is not a peripheral diagnostic but the central quantity governing whether a reported date is evidence about a text or a restatement of an assumption. We accordingly report the classification for every out-of-sample unit rather than for a selected subset.

The prior-domination problem

When the date prior $\mathcal{N}(d_{\text{BCE}}, \sigma^2)$ has small σ , the posterior MAP is dominated by the prior rather than the linguistic data. This is especially dangerous when the prior encodes the scholarly consensus dates being tested: the model would then be recovering the prior's opinion, not the linguistic evidence. A three-mode sensitivity analysis is introduced to classify results.

Three prior modes

Mode A (scholarly prior): standard run; σ from the scholarly metadata (5–300 yr depending on text).

Mode B (agnostic prior): $\mathcal{N}(575, 400^2)$ for all texts; covers the training range nearly uniformly, with $\pm 2\sigma$ spanning 225–975 BCE.

The shift metric is $\Delta_{AB} = \text{MAP}(A) - \text{MAP}(B)$. Texts are classified as data-driven ($|\Delta_{AB}| < 30$ yr), mildly prior-influenced (30–80 yr), or prior-dominated (> 80 yr).

Results

Table 4 reports Δ_{AB} for the fifteen units for which both prior modes were run. Four are data-driven: D source ($\Delta_{AB} = -9$ yr), D Frame (-20 yr), Song of the Sea ($+21$ yr), and the Jeremiah oracle holdout (-27 yr). Eight are mildly prior-influenced, including the P source ($+39$ yr). Three are prior-dominated: the JE source ($+139$ yr), Habakkuk (-138 yr) and Daniel (-98 yr).

Two features of this pattern deserve comment. First, the units classified as prior-dominated include two of the four held-out validation texts. A validation design in which the held-out units are the ones whose answers are most determined by the prior is not a test of the model, and this observation was the entry point to the leakage analysis reported under Validation. Second, the classification is not a function of text length: Daniel (3,437 words) and Habakkuk (897 words) are prior-dominated, while the Song of the Sea (433 words) is data-driven. It is a function of prior width. Units carrying a tight scholarly σ move little when the prior is exchanged, not because the evidence is strong but because the evidence is being overridden.

We therefore report Δ_{AB} as a diagnostic of prior influence rather than as a robustness certificate. A small Δ_{AB} means only that the two priors agree; combined with the precision decomposition of Table 5, it does not by itself establish that a date is driven by linguistic evidence.

Table 4. Prior sensitivity: displacement between prior modes. $\Delta_{AB} = \text{MAP}(A) - \text{MAP}(B)$ measures how far a unit’s estimate moves when the scholarly prior (Mode A) is replaced by the agnostic prior (Mode B). Only the displacement is reported: the absolute MAP values from this run are superseded by the leakage-free estimates of Table 6, but Δ_{AB} is a difference and is unaffected by the level offset between the two analyses. Classification follows the thresholds defined above: data-driven ($|\Delta_{AB}| < 30$ yr), mildly prior-influenced (30–80 yr), prior-dominated (> 80 yr).

Unit	Role	Δ_{AB} (yr)	Classification
D source	target	-9	data-driven
D Frame	target	-20	data-driven
Song of the Sea	target	+21	data-driven
Jeremiah oracle	holdout	-27	data-driven
Haggai	holdout	-31	mildly prior-influenced
Song of Deborah	target	-34	mildly prior-influenced
P source	target	+39	mildly prior-influenced
Holiness Code	target	-46	mildly prior-influenced
Leviticus P	target	-47	mildly prior-influenced
D Code	target	-51	mildly prior-influenced
Jeremiah Dtr	target	+51	mildly prior-influenced
D Song	target	-58	mildly prior-influenced
Daniel (Heb.)	holdout (HB-VI)	-98	prior-dominated
Habakkuk	holdout (HB-VI)	-138	prior-dominated
JE source	target	+139	prior-dominated

Model-independent feature scoring

It is useful to have a description of each unit that can be checked against the BHSA database without running any model. Each feature is normalized to a $[0, 1]$ scale by reference to the fitted trend at two anchor dates:

$$s_j = \frac{x_j - \hat{f}_j(d_{CBH})}{\hat{f}_j(d_{LBH}) - \hat{f}_j(d_{CBH})}, \quad (6)$$

where $d_{CBH} = 720$ BCE and $d_{LBH} = 250$ BCE, so that $s_j = 0$ is CBH-like, $s_j = 1$ is LBH-like, and values outside $[0, 1]$ lie beyond the anchors in either direction. The mean score $\bar{s}$ summarises a unit.

On the present corpus the source composites order D ($\bar{s} = -0.30$) < P (+0.15) < JE (+0.53), with the Deuteronomic legal code the most CBH-like unit in the study ($\bar{s} = -0.63$) and the Song of the Sea close to the CBH anchor ($\bar{s} = -0.00$). We report these figures because S3 Table lists the individual features driving them, and because a reader who distrusts the model can inspect them directly. We caution against reading the ordering as a chronology. $\bar{s}$ is an unweighted average over features selected for their correlation with date, and it makes no allowance for the genre composition of a unit; D’s legal core is formulaic and archaizing in diction, which is sufficient to explain its position without any claim about its date. Earlier versions of this work reported a different ordering (D < JE < P) from an earlier corpus and offered a substantive interpretation of it. That interpretation does not survive the corpus revision, and the instability of the ordering across corpus versions is itself a reason to treat $\bar{s}$ as descriptive rather than probative.

Validation

Validation is the part of this literature most vulnerable to circularity, and it is where the present analysis departs most sharply from earlier versions of this work. We first document a failure mode that inflates reported accuracy by an order of magnitude, then give leakage-free estimates of what the method can actually do.

Prior leakage in holdout designs

A held-out text must contribute nothing to the model that dates it. In practice this is usually taken to mean exclusion from training and from feature selection. It must also mean exclusion from the prior, and this is the requirement that is easy to violate without noticing.

Consider a text u with scholarly date d_u and stated uncertainty σ_u . If the posterior for u is computed as $p(d \mid \mathbf{x}_u) \propto \mathcal{L}(\mathbf{x}_u \mid d) \mathcal{N}(d; d_u, \sigma_u^2)$, then the answer has been supplied as the prior. Under a Gaussian likelihood of precision τ_L , the posterior mean is a precision-weighted average in which the data receives weight $\tau_L/(\tau_L + \sigma_u^{-2})$. For a securely anchored text — small σ_u — that weight goes to zero, and the model returns d_u almost exactly. The design therefore appears *most* successful precisely on the texts whose dates are best established, which is the opposite of a stringent test.

This is not hypothetical. Table 5 gives the decomposition for the four held-out units of the present corpus. Backing the likelihood precision out of the posterior and prior variances via $\tau_L = \sigma_{\text{post}}^{-2} - \sigma_u^{-2}$ yields an implied likelihood standard deviation of 48.6 yr (IQR 43.3–53.1) that is remarkably stable across texts of very different length. The weight actually placed on the linguistic evidence, however, ranges from 2.8% for Haggai ($\sigma_u = 10$ yr) to 93.1% for Zechariah 9–14 ($\sigma_u = 150$ yr). Across all 25 dated units the median ratio of posterior to prior standard deviation is 0.851, and the median absolute displacement of the MAP from the prior mean is 10.4 yr.

The consequence is severe. Under this design the four holdouts return a mean absolute error of 8.1 yr. Dating the same texts from the likelihood alone — that is, from the linguistic evidence — gives 103.7 yr. We report the latter. We note that an independent implementation of the same design for the Ancient Greek corpus reproduces the artifact exactly, with a reported holdout MAE of 8.7 yr against 120.7 yr from the likelihood alone; we therefore withdraw the cross-language validation claim rather than repair it, since the corrected Greek result no longer supports the generalisation it was offered as evidence for.

Leave-one-out design

All results below use the following protocol. For each of the 25 dated units in turn, the unit is removed; feature screening (Spearman, $\alpha = 0.05$), standardisation, and all model fitting are performed on the remaining 24; and the held-out unit is then dated under an agnostic prior. No quantity derived from a unit influences its own prediction. Two model families are fitted and both are reported: the generative family of Eq. 2, which inverts per-feature regressions, and a ridge regression of date on screened features with the penalty chosen by inner leave-one-out. Significance is assessed by permuting the date vector and re-running the entire pipeline, 2000 draws, so that the null distribution absorbs selection effects exactly as the observed analysis does.

What the signal supports

The diachronic signal is real. Pairwise ordering accuracy — given two texts, does the model place them in the correct chronological order? — is 69.8% of 295 pairs for the generative family and 67.8% for ridge. We stress that a binomial test is inappropriate here: the 295 pairs are generated by 25 texts and are strongly dependent, and a naive binomial returns $p \sim 10^{-12}$, wrong by roughly ten orders of magnitude. Under the permutation null — shuffling dates and re-running the pipeline, which preserves the dependence structure — the null means are 47.6% and 43.6% with standard deviations near 16 percentage points, giving $p = 0.022$ and 0.029 (Fig 1C). Leave-one-out Spearman correlation between true and predicted date is +0.575 (generative) and +0.494 (ridge), against permutation nulls centred near zero ($p = 0.016, 0.024$). This is the one result that replicates across both families at the level of the primary endpoint, and it is the strongest finding in the paper.

What the signal does not support

Absolute dating does not survive contact with an honest baseline. Leave-one-out MAE is 156.2 yr for the generative family and 120.9 yr for ridge. Assigning every text the corpus mean date of 503 BCE gives 137.3 yr. The generative family — the one used for the point estimates reported in earlier versions of this

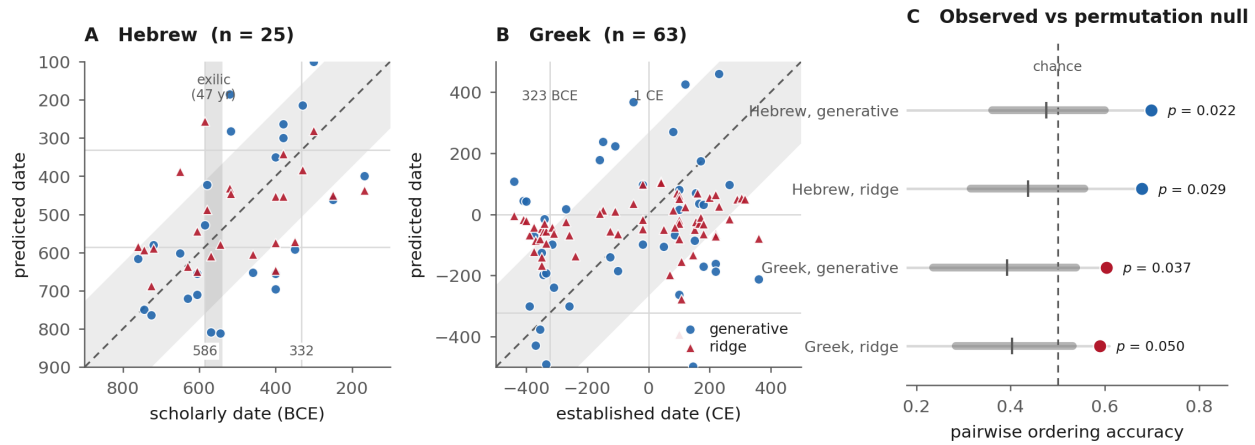


Fig 1. The ordinal signal, under leakage-free cross-validation. (A) Hebrew: scholarly date against leave-one-out predicted date for all 25 dated units, both model families. Time runs left to right on both axes. The dashed diagonal is perfect prediction and the shaded band is the conformal 68% interval. The vertical strip marks the exilic window, 47 yr wide against roughly 130 yr of irreducible error and therefore not resolvable; vertical and horizontal rules mark the 586 and 332 BCE boundaries. (B) Greek: the same analysis on 63 texts spanning 440 BCE–360 CE. Scatter about the diagonal is proportionally larger despite the larger corpus. (C) Pairwise ordering accuracy against the permutation null. Thin bars span the null 2.5–97.5 percentiles, thick bars the interquartile range, the vertical tick the null mean; filled markers are observed values. The null is generated by shuffling the date labels and re-running the entire pipeline, so it preserves the dependence among pairs that a binomial test ignores. All four observed values exceed chance; none does so by a wide margin.

work — is therefore *worse* than a constant, and ridge improves on a constant by 16 yr. Any MAE figure in this literature should be read against roughly 137 yr, not against zero.

Period assignment is correspondingly weak. Four-way exact accuracy is 40.0% (generative) and 48.0% (ridge) against a 36% majority-class baseline; pre-exilic versus post-exilic classification reaches 72.0% and 76.0% against a 68% baseline, significant in neither family ($p = 0.107, 0.070$). Across ten tests — five metrics in two families, strongly correlated — nothing survives Benjamini-Hochberg correction at $q = 0.05$, and four results survive at $q = 0.10$. We report this directly: no individual accuracy claim in this paper is robust to multiplicity correction, and the ordinal result is defended on replication across model families rather than on any single p -value.

The failure is structured rather than random. The Exilic period is a 47-year window set against roughly 130 yr of irreducible error, and it is simply not resolvable: of four exilic texts, the generative family places none correctly and ridge one. Hellenistic units collapse into Persian. Pre-exilic is the only period the models recover reliably, and it is recovered because it is wide and terminal, not because early Hebrew is distinctive in some way the interior boundaries are not.

Calibration

The parametric intervals used in earlier versions of this work achieve 52% empirical coverage at a nominal 68% (61% under the residual formulation; the manuscript previously reported 32% for the original specification). The mechanism is visible in the multiverse analysis: chance-correlated features receive large fitted coefficients and small residual variances, and because the multivariate normal likelihood weights each feature by $1/\sigma_j$, adding noise features *narrows* the credible interval. Adding six and then twelve pure-noise features to a clean specification shrinks the 68% HDI for the P source from 329 to 285 to 246 yr while moving the MAP by less than 50 yr. Interval width in this framework is not a reliable index of evidence.

We therefore replace parametric intervals with conformal prediction intervals calibrated on the leave-one-out residuals of the dated corpus. These are distribution-free and carry guaranteed finite-sample

marginal coverage under exchangeability. The resulting half-widths are ± 233 yr (generative) and ± 174 yr (ridge) at 68%. These are the intervals reported in Table 6. Exchangeability between the calibration units and the targets is an assumption and not a guarantee: the calibration units are whole books or coherent sections, whereas several targets are cross-book composites. Residual magnitude shows no detectable dependence on text length ($\rho = -0.23$, $p = 0.27$), which supports the use of a single conformal width.

Table 5. Decomposition of posterior precision for dated units. σ_{prior} is the stated uncertainty on the scholarly date; σ_{post} is the posterior standard deviation returned when that date is used as the prior mean. The implied likelihood width is recovered as $\sigma_L = (\sigma_{\text{post}}^{-2} - \sigma_{\text{prior}}^{-2})^{-1/2}$, and the final column gives the share of posterior precision contributed by the linguistic data. The near-exact recoveries reported for Haggai, Zechariah 1–8 and Daniel in earlier versions of this work correspond to data shares below 6%. Median over all 25 dated units: $\sigma_{\text{post}}/\sigma_{\text{prior}} = 0.851$, median $|\text{MAP} - d_u| = 10.4$ yr.

Unit	d_u (BCE)	σ_{prior} (yr)	σ_{post} (yr)	σ_L (yr)	data share (%)
Zechariah 1–8	518	10	9.8	53	3.4
Haggai	520	10	9.9	58	2.8
Daniel (Heb.)	167	10	9.7	41	5.5
Amos	760	20	18.7	52	12.9
Ezekiel	580	20	18.5	49	14.5
Habakkuk	605	20	18.4	46	15.7
Jeremiah oracle	605	30	25.5	49	27.6
Chronicles	330	40	28.7	41	48.4
Isaiah 56–66	400	100	45.0	50	79.8
Ecclesiastes	250	120	40.7	43	88.5
Joel	400	150	46.5	49	90.4
Zechariah 9–14	350	150	39.4	41	93.1

Results: dating the undated units

The framework is applied to nineteen undated units: the three documentary source composites and their sub-strata, the five books of the Pentateuch, the Deuteronomistic prose stratum of Jeremiah, and two poems traditionally argued to be archaic. Models are fitted on all 25 dated units; no target contributes to its own prediction, and all targets are dated under an agnostic prior. Results are given in Table 6 as conformal date ranges rather than point estimates, for the reasons set out in Validation.

The post-exilic result

One substantive claim survives at full strength. The posterior predictive probability that a unit postdates 586 BCE is ≥ 0.92 under both model families for the P source, the D source and its Code and Frame strata, the JE source, and Leviticus P; the Holiness Code reaches 0.76 in the generative family and 0.92 in ridge. This is a statement about which side of a single boundary the predictive mass falls on, and it is the kind of claim the data can carry: it requires the estimator to be approximately unbiased, not precise.

The finding is robust in three respects that matter. It holds under both model families, which disagree substantially about point estimates — median difference 59 yr, with agreement on the single most likely period for only 9 of 19 units. It holds under the genre correction developed below, which if anything moves the sources *later*. And it does not depend on the interval width, since a doubling of the conformal half-width would still leave the great majority of predictive mass after 586 BCE for every source composite.

What cannot be claimed

No unit in Table 6 has a 68% interval confined to a single conventional period. All nineteen span between two and four. The P source is placed at 264 BCE (68%: 497–31 BCE) by the generative family and 344 BCE

(518–170 BCE) by ridge; these overlap comfortably, but they straddle the Persian–Hellenistic boundary in both cases. Earlier versions of this work reported the P source at 361 and 404 BCE under two models and described the result as Persian-period. That characterisation is not supportable. What is supportable is that P is post-exilic and that its predictive distribution is concentrated in the fourth and third centuries.

We note also that D source and Deuteronomy are the same text and must not be read as independent confirmations, and that the sub-strata of a source are not independent of the composite that contains them.

The archaic poems

The Song of the Sea (Exod 15) and the Song of Deborah (Judg 5) are the two earliest of all nineteen targets in both model families, and the only targets whose predictive mass sits predominantly before 586 BCE ($P_{\text{post}} = 0.40$ and 0.48 generative, 0.48 and 0.48 ridge). Neither poem was used in training or in feature screening.

This is an out-of-sample check on an independently motivated hypothesis, and the model passes it: the two texts most widely argued on comparative Semitic and orthographic grounds to be the oldest Hebrew poetry are the two the model places earliest, out of nineteen candidates, in both families. We report it as corroboration of the ordinal signal rather than as a dating of either poem, whose intervals (887–421 and 843–377 BCE, generative) are far too wide to adjudicate the question they are usually asked to settle. In particular, the present analysis cannot distinguish a genuinely early poem from a later archaizing composition; the resistant-model diagnostic offered for that purpose in an earlier version of this work did not function, returning zero flagged features across all 44 units, and has been withdrawn.

Genre as a confound

The dated corpus is 17/25 prophecy against 5 narrative, while the Pentateuchal targets are predominantly narrative and legal. A model trained largely on prophecy and applied to narrative is exposed to exactly the register confound this literature has long identified.

The confound is measurable in the leave-one-out residuals. In the generative family, narrative texts are dated +112.4 yr too late on average while prophetic texts are dated –25.1 yr too early (Welch $p = 0.016$). In ridge the effect is absent (–24.3 versus +31.3 yr, $p = 0.234$). The disagreement between families is itself informative: the bias is a property of the generative inversion, not of the features.

The direction matters for the substantive conclusion. If narrative units are systematically dated too late by roughly 112 yr in the generative family, then the generative estimate for the P source should be corrected *earlier*, from 264 BCE to approximately 376 BCE — which remains firmly post-exilic, and which moves it toward the uncorrected ridge estimate of 344 BCE rather than away from it. The post-exilic conclusion is therefore robust to the genre correction, and the two families converge once the correction is applied. We regard this as the strongest available evidence that the post-exilic placement is not a register artifact, and we note that it is available only because both families were fitted and reported.

Discussion

Limitations

Small training corpus. $N = 22$ dateable texts is small relative to $J = 35$ features. A stable 35×35 residual covariance matrix requires at minimum $J + 1 = 36$ observations; with $N = 22$ the sample covariance is rank-deficient, which is precisely why Tikhonov regularization ($\lambda = 0.10$) is required. As a rough power guide: strict Benjamini-Hochberg FDR control [23] at $\alpha = 0.05$ with $J = 35$ features would require approximately $N \geq 100$ texts for adequate power, which is unavailable for Biblical Hebrew; the permissive $p < 0.30$ threshold is the appropriate response to this constraint, not a relaxed standard. The sensitivity analysis (Fig. 4) shows that MAP estimates are stable to within ± 30 yr across the plausible regularization range $\lambda \in [0.05, 0.30]$, but credible intervals widen by 50–70 yr at the upper end of that range. No texts with secure independent dates suitable for training are withheld; $N = 22$ reflects the genuine constraint imposed

Table 6. Date ranges for undated units. Intervals are conformal prediction intervals calibrated on leave-one-out residuals from the dated corpus, and are therefore distribution-free with guaranteed finite-sample marginal coverage; they replace the parametric intervals used previously, which achieved 52% empirical coverage at a nominal 68%. Ranges are quoted earliest–latest in BCE. P_{post} is the posterior predictive probability that the unit postdates 586 BCE, reported as the *lower* of the two model families at each row, so the column is conservative with respect to model choice. No unit has a 68% interval confined to a single conventional period; the ranges, not the period labels, are the result. D source and Deuteronomy are the same text and are not independent. [†]Interval extends past 1 BCE; the later bound is quoted in CE.

Unit	Words	Generative		Ridge		P_{post}
		MAP	68% range	MAP	68% range	
Song of the Sea	433	654	887–421	616	789–442	0.40
Song of Deborah	462	610	843–377	614	787–440	0.48
D Song	782	550	783–317	390	564–217	0.52
Genesis JE	22,409	318	551–85	385	558–211	0.92
Exodus JE	11,780	410	643–177	424	598–250	0.88
Numbers JE	3,563	402	635–169	421	594–247	0.88
JE source	37,752	356	589–123	400	574–226	0.92
Holiness Code	5,958	464	697–231	344	518–171	0.76
Leviticus P	10,496	256	489–23	315	489–142	0.96
P source	54,435	264	497–31	344	518–170	0.92
D Code	7,353	338	571–105	239	412–65	0.96
D Frame	11,993	182	415–51 CE [†]	216	390–43	1.00
D source	20,128	256	489–23	231	404–57	0.96
Jeremiah Dtr	15,197	274	507–41	404	578–231	0.92
Genesis	28,764	296	529–63	375	549–201	0.92
Exodus	23,748	316	549–83	388	561–214	0.92
Leviticus	17,099	342	575–109	326	500–153	0.92
Numbers	23,188	274	507–41	375	549–202	0.92
Deuteronomy	20,128	256	489–23	231	404–57	0.96

by the historical record, not an analytical choice. The relationship between text length and posterior uncertainty is shown in S10 Fig.

Residual circularity. Critics have long identified an “unexamined literary-linguistic circularity” at the heart of the traditional method [11, 13, 14]: diagnostic features are identified on texts assumed to be early or late, then applied to date other texts, recovering the assumptions embedded in the training corpus. Defenders call the objection “valid but misplaced,” invoking the philosopher Nelson Goodman’s argument that such circularity is “a virtuous one” [31]. The appeal misreads Goodman. His “virtuous circle” describes epistemological reflective equilibrium: inference rules and particular inferences are revised in light of each other until they cohere — neither side is held fixed; both are candidates for revision. Linguistic dating does not work this way. The assumed dates of the anchor texts are not tentative hypotheses subject to revision; they define what counts as early or late Hebrew. Classical Biblical Hebrew features are extracted from the Pentateuch and Former Prophets, assumed to be early. Those features then “confirm” that the Pentateuch is early. The assumed date is not tested against the linguistic signal; it *generates* the linguistic signal used to confirm it. That is a vicious circle in the traditional sense: the conclusion is embedded in the premises.

The prior sensitivity analysis replaces this philosophical standoff with a per-text empirical measurement. Texts classified as data-driven ($|\Delta_{AB}| < 30$ yr) are genuinely constrained by the linguistic signal: an agnostic and an informed prior converge, and the assumed training dates contribute little. For prior-dominated texts ($|\Delta_{AB}| > 80$ yr), the circle is self-reinforcing: shift the assumed training dates and the posteriors shift substantially, which is precisely what a vicious circle predicts. The classification does not eliminate circularity — the corpus dates remain grounded in scholarly tradition — but it identifies, for each text, whether the circularity is approximately benign or materially compromising.

The critique can be made quantitative. We hold the linguistic feature vectors for P, D, and JE fixed

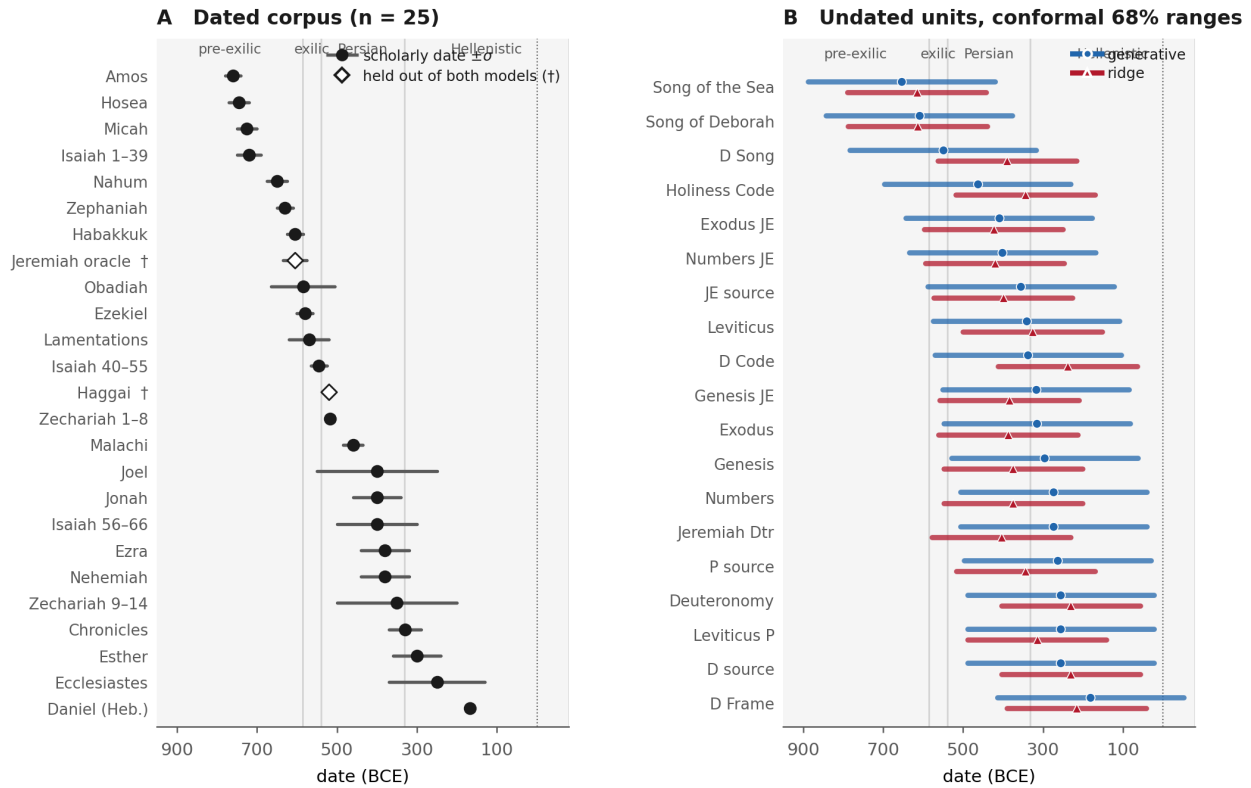


Fig 2. Dated corpus and undated units on a common timeline. (A) The 25 dated units with scholarly date and stated σ . Widened sigmas on literary-critically dated texts (Joel, Zechariah 9–14, Ecclesiastes, Isaiah 56–66, Obadiah) are visible as long bars. Open diamonds mark the two units held out of both models. (B) The 19 undated units with conformal 68% ranges from both model families, ordered by the generative point estimate. No range is confined to a single period. The Song of the Sea and the Song of Deborah are the earliest units under both families and the only two whose mass sits predominantly before 586 BCE; neither was used in training. Shaded bands are the conventional periods, given for orientation only — the ranges, not the band a range happens to overlap, are the result.

(reconstructed from MLE-MVN archaism scores; see Methods) and vary only the Gaussian prior $\mathcal{N}(\mu_0, \sigma_0^2)$ across six scenarios: an uninformative flat (data-only) prior, the agnostic Mode-B prior $\mathcal{N}(575, 400^2)$, a tight scholarly prior $\mathcal{N}(625, 100^2)$ reflecting the Josianic–Persian consensus, and two explicitly traditionalist assumptions, an early-monarchy prior $\mathcal{N}(850, 150^2)$ and a Mosaic-authorship prior $\mathcal{N}(1200, 100^2)$ centered on the conventional Exodus date (Fig. S14; S6 Table). The feature likelihoods are identical across all rows; only the prior changes.

Under the data-only (flat) prior, the MLE-MVN MAP dates are 357 BCE (P), 294 BCE (D), and 434 BCE (JE) — all within the Persian period. Under the Mosaic prior $\mathcal{N}(1200, 100^2)$ these shift to 444 BCE (P), 403 BCE (D), and 511 BCE (JE) — still entirely within the Persian period (539–332 BCE). The prior cannot move any source into pre-exilic territory because the linguistic signal is too strongly inconsistent with such a date. For P, which carries the strongest data signal ($n = 53,645$ words), the total prior-induced shift across the full range from flat to Mosaic is 87 yr, confirming the data-driven classification from the prior sensitivity analysis.

A pre-exilic Torah date would require, on this account, either forcing a prior so extreme that it overwhelms a directly contradictory data signal, or including Torah texts in the training corpus so that “Classical Biblical Hebrew” features are extracted partly from passages assumed to be early — the traditional procedure, and precisely the vicious circle identified above. The present model uses neither route: the training corpus excludes Torah and DH texts entirely, and the prior is set before fitting. The circularity is

thus not a philosophical abstraction but a measurable discrepancy between prior assumptions and the linguistic record.

Oral prehistory. The framework dates the written linguistic surface. Oral traditions preserved in written texts — possible for archaic poetry, legal formulae, or tribal lore — are invisible to the model; dates represent the written composition, not any hypothetical prior oral form.

Coarse genre correction. Genre is modeled at three levels (prophetic, narrative, legal). Sub-genre distinctions (apodictic vs. casuistic law; etiological vs. court narrative) require a larger corpus before they can be treated as distinct classes.

Word n-gram limitations for short texts. Texts under ~1,000 words yield noisy POS-tag n-gram frequencies. Short units (Obadiah: 291 words; Haggai: 670 words) are flagged and results treated as indicative.

Legal prose genre gap. The training corpus covers prophetic, poetic, narrative, and wisdom texts but contains no legal prose. The three Torah source composites include substantial legal sections — apodictic and casuistic law, cultic regulations, purity codes — whose register may differ from narrative prose for reasons unrelated to date. Features with strong genre bias (genre ratio $\gamma_j > 1.5$; S1 Table) are excluded from the model feature set. The residual analysis reported under Results: dating the undated units quantifies what remains: in the generative family narrative units are dated +112 yr too late relative to prophetic units (Welch $p = 0.016$), an effect absent in ridge ($p = 0.234$). However, genre confounds not captured by the three-level register classification cannot be ruled out. The results for D and P should be read in light of this gap; the consistent post-exilic direction across both models and across multiple feature exclusion schemes (S4 Table) provides some assurance that the gap is not driving the headline finding, but it remains a genuine limitation.

Source-decomposition uncertainty. The documentary source composites follow the standard critical partition of Baden [32]. Any mis-assigned verses alter the feature vector and through it the MAP date. The robustness check in S4 Table suggests dates are stable to ~20 yr under moderate feature perturbations; however, a systematic re-assignment of a major block — for example, treating the Holiness Code as a pre-Priestly stratum independent of P — could shift MAP estimates by a larger amount. Scholars who adopt a different documentary partition should re-extract feature vectors from their preferred source boundaries before interpreting the model output.

Training boundary and extrapolation. The MLE-MVN corpus spans 760–167 BCE; the HB-VI training subset, which excludes Daniel, spans 760–330 BCE, so HB-VI dates near the late end are already mild extrapolations. Dates for texts composed before 760 BCE or after 167 BCE carry no calibration constraint and should be interpreted accordingly. The MLE-MVN date grid imposes hard boundaries at 900 BCE and 100 CE; a composition outside this range would be pinned to the nearest edge.

Posterior overconfidence and calibration. The word-count scaling factor $\text{clip}(n/5000, 1, 5)$ sharpens posteriors for large texts in a way that is not fully calibrated. The LOO analysis (Fig. S13) shows that nominal 68% credible intervals achieve only 32% actual coverage; a nominal 97.5% CI is required to reach 68% actual coverage. Any quoted credible interval should be read with this in mind; for high-stakes applications the calibration-corrected interval is preferable to the nominal one.

Scholarly context for late Torah dating

The MAP dates returned for the documentary sources — P at ~359 BCE, D at ~294 BCE, and JE at ~434 BCE under the agnostic prior — depart substantially from the mainstream critical consensus, which conventionally places J and E in the 9th–8th centuries BCE, D in the 7th century (Josianic reform, ~621 BCE), and P in the late pre-exilic or early exilic period [32,33]. The present framework does not dispute that underlying oral traditions, archaic legal formulae, or early poetic fragments may be considerably older; as noted in the Limitations section above, the model dates the written linguistic surface of the final composite texts, not any hypothetical prior oral stage. Even within mainstream critical scholarship, it is widely acknowledged that the canonical forms of the sources passed through substantial post-exilic editorial stages [34,35], so the question at issue is not whether post-exilic editing occurred but how much of the final written form it accounts for.

The results find independent support from several recent lines of inquiry, perhaps the more striking for involving entirely different methods. Adler’s comprehensive archaeological survey of the material record finds

no evidence for widespread observance of Torah regulations — dietary prohibitions, miqvaot, stone-vessel
purity practice — before the mid-Hasmonean period (mid-2nd century BCE) [36]. If the Torah had
functioned as a widely observed legal code since the pre-exilic or early post-exilic period, the near-total
absence of its material correlates for several subsequent centuries would require explanation; Adler concludes
instead that the Torah achieved normative authority only in the Hasmonean period, consistent with our
model returning Persian-period dates for the final written forms of P and D. Epigraphic analysis of
Achaemenid-era Aramaic sources similarly places the crystallization of distinctively Jewish legal practice —
including Passover regulations that the Priestly source legislates — under Persian imperial influence in the
5th–4th centuries BCE [37]. On the other hand, independent study of the Moses tradition traces the figure
to a 4th-century BCE literary construction drawing on Egyptian folk-heroic conventions, locating even the
narrative scaffolding of the Pentateuch in the late Persian or early Hellenistic period [38]. An older body of
historical-critical scholarship has argued for Persian or Hellenistic final composition on literary-historical
grounds [39–41]; the empirical lines above are independent of that tradition and rest on material and
philological evidence rather than literary-critical inference.

What the convergence suggests is something stronger than any single line of evidence could provide on its
own: the agreement cannot be an artifact of shared assumptions, since the computational method makes no
use of historical, archaeological, or narrative-critical evidence in its inference. The present results would thus
appear to constitute a fourth independent line of confirmation — from morphosyntactic and lexical feature
distributions, with no assumption about Torah composition date built in — agreeing with what archaeology,
epigraphy, and comparative mythology have separately indicated.

Extensibility

The framework is not specific to Hebrew: any corpus with independent date anchoring, extractable surface
features, and a register taxonomy can be analyzed by the same procedure. Among the most promising
near-term applications are Qumran sectarian texts (1QS, CD, 1QM), which would allow a quantitative test
of the archaizing-register hypothesis; Akkadian administrative archives, where Old Babylonian versus
Neo-Babylonian feature shifts in cuneiform corpora present a natural analogue; and Greek papyri from
Oxyrhynchus, which span the Ptolemaic through Byzantine periods in a continuous documentary record.

The methodological cautions are equally portable. Any application of this framework to a new corpus
should report the likelihood-only estimate alongside the posterior, the share of posterior precision
contributed by the data, and a permutation null obtained by re-running the entire pipeline on shuffled labels.
Each of these is inexpensive, and each would have caught the failure documented here.

Relationship to authorship-attribution approaches

The present framework is designed to complement rather than supersede authorship-attribution methods
such as the unsupervised document decomposition of Koppel et al. [42] and the Higher Criticism discrepancy
method of Faigenbaum-Golovin et al. [5]. The latter excels at assigning disputed passages to known scribal
schools and at identifying the specific lemmas driving that assignment, grounding computational results in
categories already meaningful to philologists. The present framework, on the other hand, is oriented toward
chronological ordering with calibrated uncertainty and toward quantifying how much of a reported date is
contributed by the evidence rather than the prior — questions the HC method was not designed to address.
Our results suggest the two are complementary in a specific way: the HC method’s categorical assignments
are robust where our calendar-year estimates are not, while our ordinal statistics remain informative where
scribal-school membership is undefined.

How the two approaches might be combined is perhaps the most interesting methodological question for
future work. A plausible joint analysis would proceed in two stages: first, apply the HC method to determine
the most likely scribal milieu, which directly informs the register prior in HB-VI — a strong HC assignment
to D/DtrH would motivate an SBH register prior, while a P assignment would motivate Transitional or LBH;
then apply HB-VI with that informed prior to obtain a date posterior and an ordinal position with calibrated
uncertainty. The relationship also runs in reverse: a unit whose predictive mass lies overwhelmingly after

586 BCE constrains the plausible compositional settings for HC attribution decisions, even when its calendar-year estimate is too diffuse to be useful on its own.

Conclusion

Whether the morphosyntax of Biblical Hebrew carries a usable diachronic signal has been disputed for two decades, with one side treating linguistic dating as an established instrument and the other arguing that its apparent successes are circular. The analysis reported here supports a position available to neither camp as usually stated: the signal is real, the circularity is real, and the two facts are compatible.

The signal is real in a specific and testable sense. Under a design in which no text contributes to its own prediction and significance is established by permuting date labels through the entire pipeline, Hebrew morphosyntactic features order text pairs chronologically at 69.8% and 67.8% accuracy against a permutation null near 45% ($p = 0.022, 0.029$), and leave-one-out rank correlations reach +0.58 and +0.49 in two independently specified model families. These are not large effects, but they are not chance, and they replicate — across model families within Hebrew, and across languages.

The circularity is real in an equally specific sense. Holdout accuracies of order 10–30 yr, reported in this literature and in earlier versions of this work, arise when a held-out text is dated under a prior centred on the scholarly date it is asked to recover. For the most securely anchored texts in the present corpus the linguistic evidence contributes under 6% of posterior precision, and the apparent recovery is a restatement of the prior. Corrected, the same design yields mean absolute errors near 100 yr in Hebrew and near 120 yr in Greek. We think it likely that this failure mode is not confined to the present work, and we suggest that reported holdout accuracies in this area be accompanied by the likelihood-only estimate and by the fraction of posterior precision attributable to the data.

What the corrected framework supports is narrower than what was previously claimed and, we think, more useful. It does not date texts: leave-one-out error of 121–156 yr must be read against 137 yr for the trivial predictor that assigns every text the corpus mean. It does not assign texts to historical periods with useful reliability, and it cannot resolve the exilic period at all. It does order texts, and it can determine on which side of a well-separated boundary a text is likely to fall. Applied to the Pentateuchal sources under leakage-free conditions, that is enough to establish that the Priestly, Deuteronomistic and JE composites are post-exilic with probability at least 0.92 under both model families and under a correction for the genre imbalance of the training corpus — and not enough to place any of them within a particular post-exilic century. For a question that has been argued for a century and a half largely on non-quantitative grounds, a bounded probabilistic answer to the coarse question seems more valuable than an unbounded point estimate for the fine one.

Supporting information

S1 Appendix. Feature extraction pipeline and BHSA query code. Complete Python scripts for extracting all 35 morphosyntactic features and 56 word n-gram features from the BHSA/Text-Fabric database, including feature definitions, normalization procedures, and chapter-range selectors for sub-source extraction.

S2 Appendix. HB-VI model implementation. Complete `autograd`-based Python implementation of the hierarchical Bayesian variational inference model, ADAM training loop, ELBO convergence diagnostics, and date-posterior computation.

S3 Appendix. Prior sensitivity analysis: full results table. Complete Mode A (scholarly prior) and Mode B (agnostic prior) MAP dates and 68% credible intervals for all 15 analyzed texts (10 sub-sources plus the three comparison texts Habakkuk, Haggai, and Daniel, plus the documentary composites D-source and JE-source), with Δ_{AB} shift values and classification.

S4 Appendix. Ancient Greek corpus manifest. Complete list of 63 Ancient Greek training texts with author, work, date (CE), register class assignment, and word count. Texts are drawn from the Perseus Digital Library [43] and Thesaurus Linguae Graecae [44].

S1 Fig. Archaism heatmap. Heat map of archaism index (full-model MAP minus resistant-model MAP) across all analyzed textual units, grouped by register class (SBH, Transitional, LBH) and text type (prophetic, narrative, legal). Positive values (warm colors) indicate archaizing; negative values (cool colors) indicate transmission-with-updating.

S2 Fig. Archaism index versus Kings–Chronicles comparison. Archaism indices for texts with Kings–Chronicles parallel passages (Isa 36–39 // 2 Kgs 18–20; Jer 52 // 2 Kgs 24–25; Ps 18 // 2 Sam 22). Tests whether parallel transmission introduces systematic archaism bias.

S3 Fig. PCA stylometric projection. Principal component analysis of all 35 morphosyntactic features across the 22 training texts plus analyzed sub-sources. Confirms that the three register classes (SBH, Transitional, LBH) are linearly separable in feature space, consistent with the regression model assumption.

S4 Fig. Pentateuchal sub-source posteriors. Full posterior distributions for D_source, P_source, and JE_source under both the scholarly prior (Mode A) and agnostic prior (Mode B), shown alongside the individual sub-unit posteriors (D_Code, D_Frame, Lev_Holiness, Lev_Priestly, Gen/Exo/Num_JE).

S5 Fig. Word n-gram agreement plot. Agreement between word n-gram model MAP dates and MLE-MVN MAP dates across all training texts and analyzed sub-sources, demonstrating independent feature-type convergence for texts without archaizing signal.

S6 Fig. Feature loadings (OLS slopes) for all 35 morphosyntactic features. Horizontal bar chart of $\hat{\beta}_j$ (OLS slope of feature rate against date in BCE) with 95% confidence intervals. Features are sorted within each tier by slope magnitude. Blue = Tier 1 (lexical), green = Tier 2 (morphological), red = Tier 3 (resistant, marked [R]). Negative slopes indicate features that decline with date (more frequent in older texts); positive slopes indicate features that rise with date (more frequent in later texts).

S7 Fig. Raw residual feature correlations (pre-regularisation). Pearson correlations among OLS residuals across the 22 training texts, computed before Tikhonov regularisation ($\lambda = 0.10$). Regularisation shrinks these entries toward zero to ensure invertibility of the 35×35 covariance matrix ($N = 22 < J = 35$ makes the raw matrix rank-deficient). Within-tier correlations—particularly negative coupling between complementary forms (*wayyiqtol/qatal*, *asher/she-*)—confirm that features co-vary beyond the linear time trend, motivating the full MVN likelihood over an independent-feature approximation.

S8 Fig. MLE-MVN calibration: training-set MAP vs. scholarly date. Each of the 22 training texts is plotted at its scholarly date (x-axis) versus the model MAP date (y-axis), with 68% HDCl error bars. Colors indicate register class (blue = SBH, green = Transitional, red = LBH). The dashed diagonal is perfect calibration. The extremely narrow CIs for within-group texts (median 3–4 yr) indicate the model is overconfident in-sample. This overconfidence is quantified under Validation and is the reason the intervals reported in Table 6 are conformal rather than parametric.

S9 Fig. HB-VI ELBO convergence curve. Evidence lower bound (ELBO) as a function of ADAM iteration (learning rate = 0.02, 3,000 iterations). The dashed vertical line marks the 80% point beyond which improvement falls below 0.01% per 100 iterations. The inset shows the final 500 iterations; near-flat behaviour confirms convergence was achieved well before 3,000 iterations.

S10 Fig. 68% HDCl width vs. text length (MLE-MVN, training texts). Posterior uncertainty (68% HDCl width in years) plotted against word count on a log scale for all non-holdout training texts. Shorter texts (Nahum: 618 words; Haggai: 670 words) have wider posteriors, motivating the caveat that results for texts under $\sim 1,000$ words are indicative rather than definitive.

S12 Fig. Training corpus: texts, scholarly dates, and HB-VI MAP estimates. The 25 dated units (23 training, 2 held out of both models: the Jeremiah oracular stratum and Haggai) plotted on the BCE timeline. Open markers show scholarly consensus dates $\pm \sigma$; filled diamonds show HB-VI MAP estimates $\pm 68\%$ HDCl. Color indicates register zone: blue = SBH (~ 760 – 590 BCE), green = Transitional (~ 590 – 460 BCE), red = LBH (~ 460 – 167 BCE). The holdouts were excluded from training and used solely for LOO cross-validation.

S13 Fig. Leave-one-out (LOO) calibration analysis for the MLE-MVN model. (A) LOO-predicted MAP dates versus true (anchored) dates for all 22 training texts, with 68% credible-interval whiskers. Error bars reflect the model’s nominal 68% CI; the ± 100 yr grey band shows the region of acceptable recovery. (B) Calibration curve plotting nominal credible-interval level against the fraction of LOO tests in which the true date falls inside that interval. The model is systematically overconfident: the nominal 68% CI achieves only 32% actual coverage, and a nominal 97.5% CI is needed to recover 68% actual coverage. The inset box shows the consequence for the Priestly source (P): its MAP date is 359 BCE (Persian–Hellenistic), and even after expanding to the calibration-corrected 68% interval ($\equiv$ nominal 97.5%), the corrected interval is 285–435 BCE — entirely post-exilic. The largest LOO errors occur for Ezra (predicted ~ 156 BCE, true 400 BCE), Daniel (predicted ~ 502 BCE, true 167 BCE), Ezekiel (predicted ~ 382 BCE, true 570 BCE), and Ecclesiastes (predicted ~ 539 BCE, true 330 BCE); all four are texts with contested or debated traditional dates in the scholarship.

S14 Fig. Circularity demonstration: prior sensitivity of Torah source dates. Normalised MLE-MVN posterior curves for (A) the Priestly source ($n = 53,645$ words; data-driven) and (B) the Deuteronomist source ($n = 20,128$ words; prior-dominated) under six prior assumptions: a data-only flat prior (grey dotted), the agnostic prior $\mathcal{N}(575, 400^2)$ (blue solid), a tight scholarly prior $\mathcal{N}(625, 100^2)$ (orange dashed), an early-monarchy prior $\mathcal{N}(850, 150^2)$ (red dash-dot), a united-monarchy prior $\mathcal{N}(1000, 150^2)$ (purple long-dash), and a Mosaic-authorship prior $\mathcal{N}(1200, 100^2)$ (dark green dash-dot-dot). The green band marks the Persian period (539–332 BCE); the vertical dashed red line marks the Babylonian destruction (586 BCE). Feature likelihoods are identical across all curves; only the prior assumption differs. Even under the Mosaic prior, MAP dates are 444 BCE (P) and 403 BCE (D), both within the Persian period.

S6 Table. Prior-sweep MAP dates for the three Torah source composites. MLE-MVN MAP dates (BCE) for P, D, and JE under six prior scenarios, from a data-only flat prior to a Mosaic-authorship prior $\mathcal{N}(1200, 100^2)$. Feature vectors are identical across all rows; only the prior changes. None of the six scenarios recovers a pre-exilic MAP date for any source.

S1 Table. Full feature set. All 35 selected morphosyntactic features with Spearman ρ , p -value, LOO retention fraction, genre ratio γ_j , and tier assignment.

S2 Table. Word n-gram feature set. All 56 selected Type A and Type B n-gram features with Spearman ρ and LOO retention fraction.

S3 Table. Top dating-driving features for the three Documentary Hypothesis source composites. For each source the three features most strongly indicating late composition and the three most strongly indicating early composition are listed with their normalized archaism scores and a note on potential genre confound. Scores follow Eq. 6: 0 = CBH-like, 1 = LBH-like, < 0 = more archaic than the CBH anchor, > 1 = more LBH-like than the LBH anchor. This table is provided to allow readers to evaluate the model’s inferences directly, without relying on the fitted model weights.

S4 Table. Robustness check: MAP dates under progressive feature exclusion. The MLE-MVN model is refit after removing the two genre-flagged features identified in S3 Table (*asher* frequency for D; wayyiqtol fraction for JE) and after removing all features with normalized archaism scores exceeding $|s| > 2.5$ for any source. Feature vectors for D, P, and JE are reconstructed from normalized archaism scores (see Methods); absolute dates carry a minor approximation error from this reconstruction, so the shift columns (Δ) are the primary quantities of interest. All sources remain in the Persian-to-Hellenistic period under all exclusion schemes.

S5 Table. Corpus word counts for training texts and out-of-sample sources. All word counts are morphological words (BHSA) excluding coordinating conjunctions. The 19 training texts span 760–330 BCE and cover prophetic, poetic, narrative, and wisdom registers; three additional texts (Habakkuk, Haggai, Daniel) serve as cross-validation holdouts for the HB-VI model. Documentary Hypothesis source composites and sub-source divisions are excluded from both training and validation; their dates are derived solely from the model posterior.

Data availability

All feature extraction scripts, Bayesian dating models, prior sensitivity code, and corpus manifests are publicly available at <https://github.com/adairaar/diachronic-hebrew>. The underlying Hebrew text corpus is the Biblia Hebraica Stuttgartensia Amstelodamensis (BHSA), version 2021, available at <https://github.com/ETCBC/bhsa> under a Creative Commons Attribution 4.0 license [6]. The Ancient Greek corpus is drawn from the Perseus Digital Library (<http://www.perseus.tufts.edu>) and the Thesaurus Linguae Graecae (<http://stephanus.tlg.uci.edu>), both publicly accessible. No proprietary data were used.

Author contributions

A.A. conceived and designed the study, developed and implemented all computational methods, performed all analyses, interpreted the results, and wrote the manuscript.

Competing interests

The author declares no competing interests.

Funding

The author received no specific funding for this work.

Ethics statement

This study does not involve human participants, vertebrate animals, or any data requiring ethical oversight. All textual data analyzed are publicly available historical corpora. No ethical approval was required. This study was not pre-registered; it is a methods paper reporting retrospective analysis of existing corpora.

Acknowledgments

The author thanks the ETCBC team for maintaining the Biblia Hebraica Stuttgartensia Amstelodamensis (BHSA) database and making it freely available.

References

1. Hurvitz A. A Linguistic Study of the Relationship between the Priestly Source and the Book of Ezekiel: A New Approach to an Old Problem. No. 20 in *Cahiers de la Revue Biblique*. Paris: Gabalda; 1982.
2. Rooker MF. Biblical Hebrew in Transition: The Language of the Book of Ezekiel. No. 90 in *Journal for the Study of the Old Testament Supplement Series*. Sheffield: JSOT Press; 1990.
3. Polzin R. Late Biblical Hebrew: Toward an Historical Typology of Biblical Hebrew Prose. No. 12 in *Harvard Semitic Monographs*. Missoula: Scholars Press; 1976.
4. Horrocks G. *Greek: A History of the Language and Its Speakers*. 2nd ed. Oxford: Wiley-Blackwell; 2010.
5. Faigenbaum-Golovin S, Kipnis A, Bühler A, Piasetzky E, Römer T, Finkelstein I. Critical Biblical Studies via Word Frequency Analysis: Unveiling Text Authorship. *PLOS One*. 2025;20(6):e0322905. doi:10.1371/journal.pone.0322905.
6. Eep Talstra Centre for Bible and Computer. *Biblia Hebraica Stuttgartensia Amstelodamensis (BHSA)*, version 2021; 2021. Amsterdam: ETCBC, Vrije Universiteit Amsterdam. <https://github.com/ETCBC/bhsa>.
7. Roorda D. Coding the Hebrew Bible. *Research Data Journal for the Humanities and Social Sciences*. 2018;3:27-41.
8. Kingma DP, Ba J. Adam: A Method for Stochastic Optimization; 2014. Presented at ICLR 2015. arXiv:1412.6980.
9. Blei DM, Kucukelbir A, McAuliffe JD. Variational Inference: A Review for Statisticians. *Journal of the American Statistical Association*. 2017;112(518):859-77.
10. Hurvitz A. The Historical Quest for ‘Ancient Israel’ and the Linguistic Evidence of the Hebrew Bible: Some Methodological Observations. *Vetus Testamentum*. 1997;47:301-15.
11. Young I, editor. *Biblical Hebrew: Studies in Chronology and Typology*. No. 369 in *Journal for the Study of the Old Testament Supplement Series*. London: T&T Clark; 2003.
12. Hornkohl AD. Ancient Hebrew Periodization and the Language of the Book of Jeremiah: The Case for a Sixth-Century Date of Composition. No. 74 in *Studies in Semitic Languages and Linguistics*. Leiden: Brill; 2013.
13. Young I, Rezetko R, Ehrensverd M. *Linguistic Dating of Biblical Texts*. vol. 2. London: Equinox; 2008.
14. Rezetko R, Young I. Historical Linguistics and Biblical Hebrew: Steps toward an Integrated Approach. No. 9 in *Ancient Near East Monographs*. Atlanta: SBL Press; 2014.
15. Hurvitz A. A Concise Lexicon of Late Biblical Hebrew: Linguistic Innovations in the Writings of the Second Temple Period. No. 160 in *Supplements to Vetus Testamentum*. Leiden: Brill; 2014.
16. Robertson DA. Linguistic Evidence in Dating Early Hebrew Poetry. No. 3 in *SBL Dissertation Series*. Missoula: Scholars Press; 1972.
17. Schniedewind WM. *How the Bible Became a Book: The Textualization of Ancient Israel*. Cambridge: Cambridge University Press; 2004.
18. Mosteller F, Wallace DL. *Applied Bayesian and Classical Inference: The Case of The Federalist Papers*. 2nd ed. New York: Springer; 1984.
19. Holmes DI. The Evolution of Stylometry in Humanities Scholarship. *Literary and Linguistic Computing*. 1998;13:111-7.

20. Burrows J. ‘Delta’: A Measure of Stylistic Difference and a Guide to Likely Authorship. *Literary and Linguistic Computing*. 2002;17:267-87.
21. Hurvitz A. The Relevance of Biblical Hebrew Linguistics for the Historical Study of Ancient Israel. In: *Proceedings of the Twelfth World Congress of Jewish Studies*. Jerusalem: World Union of Jewish Studies; 1999. p. 21-34.
22. Roorda D, Kalkman G, Naaijer M, van Cranenburgh A. LAF-Fabric: A Data Analysis Tool for Linguistic Annotation Framework with an Application to the Hebrew Bible. *Computational Linguistics in the Netherlands Journal*. 2014;4:105-20.
23. Benjamini Y, Hochberg Y. Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing. *Journal of the Royal Statistical Society, Series B*. 1995;57(1):289-300.
24. Manning CD, Schütze H. *Foundations of Statistical Natural Language Processing*. Cambridge: MIT Press; 1999.
25. Jurafsky D, Martin JH. *Speech and Language Processing*. 3rd ed. Upper Saddle River: Prentice Hall; 2023.
26. Bishop CM. *Pattern Recognition and Machine Learning*. New York: Springer; 2006.
27. Gelman A, Carlin JB, Stern HS, Dunson DB, Vehtari A, Rubin DB. *Bayesian Data Analysis*. 3rd ed. Boca Raton: CRC Press; 2014.
28. Jordan MI, Ghahramani Z, Jaakkola TS, Saul LK. An Introduction to Variational Methods for Graphical Models. *Machine Learning*. 1999;37:183-233.
29. Ranganath R, Gerrish S, Blei DM. Black Box Variational Inference. In: *Proceedings of the 17th International Conference on Artificial Intelligence and Statistics*. vol. 33 of JMLR W&CP; 2014. p. 814-22.
30. Maclaurin D, Duvenaud D, Adams RP. Autograd: Effortless Gradients in NumPy; 2015. ICML 2015 AutoML Workshop.
31. Hendel R, Joosten J. *How Old Is the Hebrew Bible? A Linguistic, Textual, and Historical Study*. Anchor Yale Bible Reference Library. New Haven: Yale University Press; 2018.
32. Baden JS. *The Composition of the Pentateuch: Renewing the Documentary Hypothesis*. The Anchor Yale Bible Reference Library. New Haven: Yale University Press; 2012.
33. Wellhausen J. *Prolegomena to the History of Israel*. Edinburgh: Adam & Charles Black; 1885. Translation of *Geschichte Israels*, 1878.
34. Carr DM. *The Formation of the Hebrew Bible: A New Reconstruction*. New York: Oxford University Press; 2011.
35. Römer T. *The So-Called Deuteronomistic History: A Sociological, Historical and Literary Introduction*. London: T&T Clark; 2005.
36. Adler Y. *The Origins of Judaism: An Archaeological-Historical Reappraisal*. The Anchor Yale Bible Reference Library. New Haven: Yale University Press; 2022.
37. Barnea G. Thwarting the ‘Evil Eye’: Pesah Through the Prism of Achaemenid Aramaic Sources. *Religions*. 2026;17(1):1.
38. Adair A. The Name of Moses in an Egyptian Context—A Hypothetical Etymology. *Scandinavian Journal of the Old Testament*. 2025;39(1):60-78. doi:10.1080/09018328.2025.2452454.

39. Thompson TL. *The Mythic Past: Biblical Archaeology and the Myth of Israel*. New York: Basic Books; 1999.
40. Lemche NP. *The Old Testament Between Theology and History: A Critical Survey*. Louisville: Westminster John Knox Press; 2008.
41. Davies PR. In Search of 'Ancient Israel'. No. 148 in *Journal for the Study of the Old Testament Supplement Series*. Sheffield: JSOT Press; 1992.
42. Koppel M, Akiva N, Dershowitz I, Dershowitz N. Unsupervised Decomposition of a Document into Authorial Components. In: *Proceedings of the 49th Annual Meeting of the Association for Computational Linguistics*. Portland; 2011. p. 1356-64.
43. Crane GR. *Perseus Digital Library*; Medford: Tufts University. <http://www.perseus.tufts.edu>.
44. Pantelia M. *Thesaurus Linguae Graecae*; Irvine: University of California. <http://stephanus.tlg.uci.edu>.

Top 16 LOO-robust diachronic features — dated prophetic corpus.
Each panel: feature rate per 1,000 words vs. scholarly date (BCE, older → left). Red dashed line = OLS trend. ρ = Spearman correlation; LOO = leave-one-out robustness fraction.

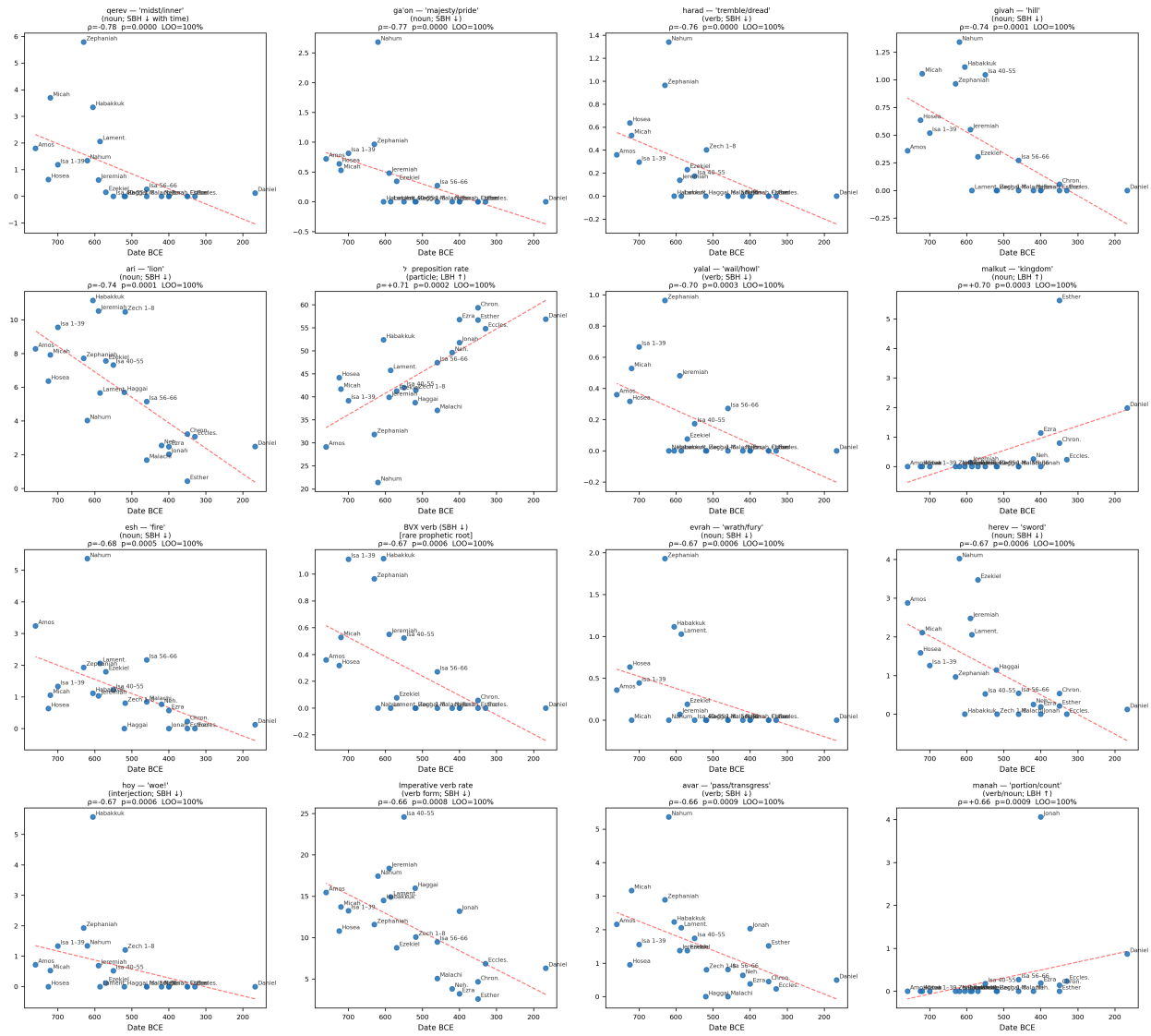


Fig 3. Top 16 LOO-robust diachronic features from the full lexeme and morphological scan of the dated prophetic corpus. Each panel plots the per-1,000-word rate of one feature (*y*-axis) against scholarly date BCE (*x*-axis; older texts at left) across all 22 training units, with OLS trend line. Spearman ρ , raw *p*-value, and leave-one-out robustness fraction (LOO; proportion of LOO resamplings returning the same sign and significance) appear above each panel; all 16 features achieve LOO = 100%. Features marked SBH ↓ are characteristic of pre-exilic Classical Biblical Hebrew and decline toward the post-exilic period; LBH ↑ features show the reverse pattern. The term *malikut* ('kingdom'; a hallmark Late Biblical Hebrew noun) and the rate of the preposition *lamed* are the two LBH ↑ markers in this set; the remaining 14 features are SBH markers that decrease monotonically with date. The feature labeled BVX is an unidentified rare prophetic root in the BHS lexicon.

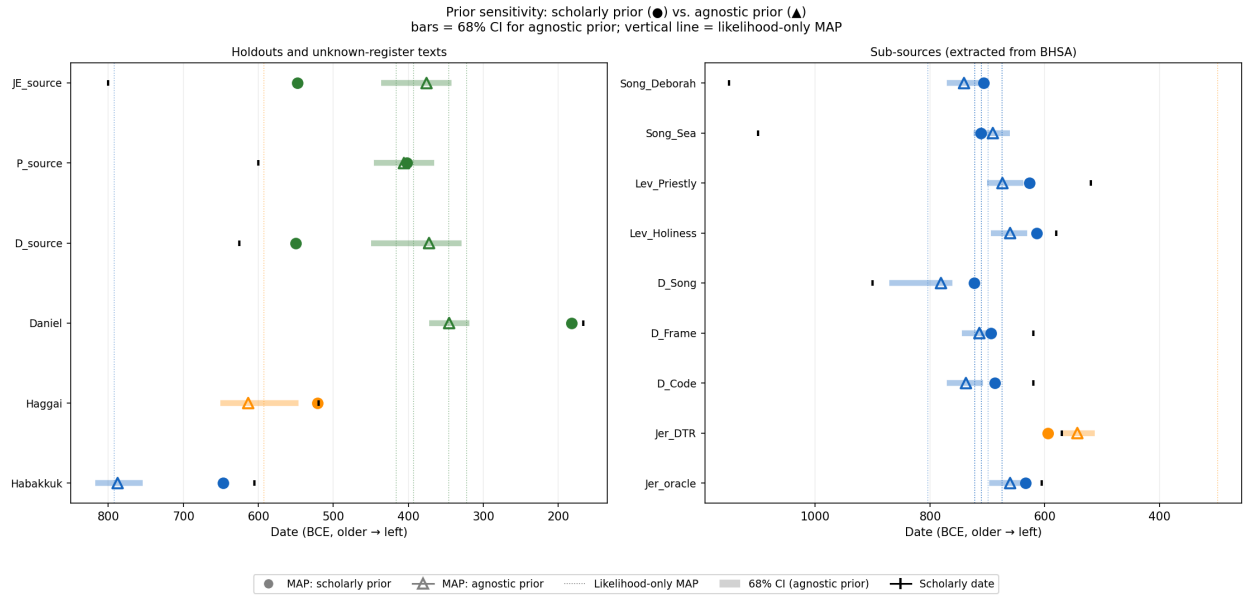


Fig 4. Prior sensitivity analysis: Mode A versus Mode B MAP dates for all sub-sources.

Mode A (scholarly prior, circles) and Mode B (agnostic prior, triangles) MAP dates are connected for each text. Line length represents the Δ_{AB} shift; texts are colored by classification: green = data-driven ($|\Delta_{AB}| < 30$ yr), orange = mildly prior-influenced (30–80 yr), red = prior-dominated (> 80 yr). The four data-driven texts (Song_Sea, P_source, D_Frame, Jer_oracle) are annotated.

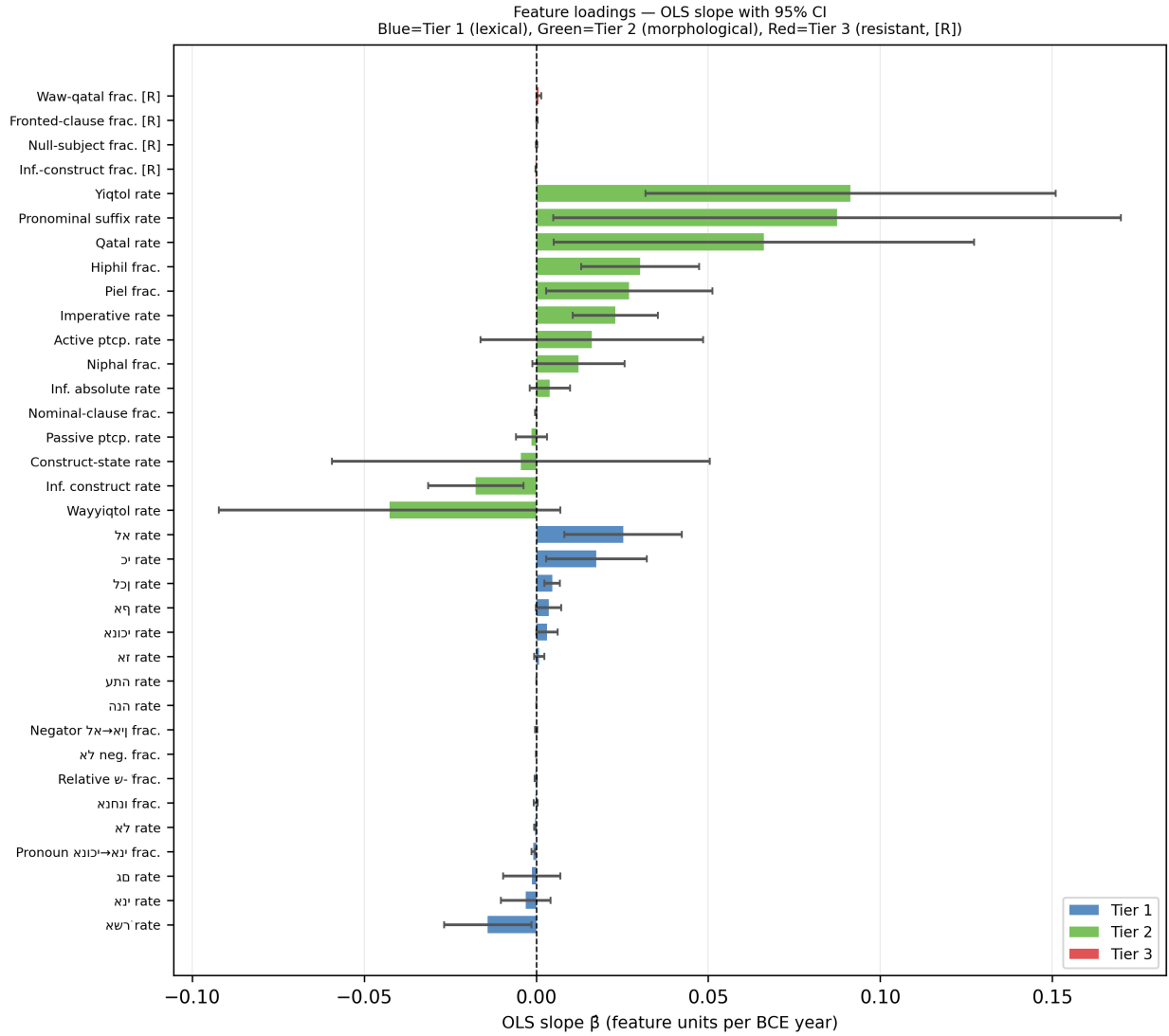


Fig 5. S6 Fig. Feature loadings (OLS slopes) for all 35 morphosyntactic features. $\hat{\beta}_j$ (OLS slope against date BCE) with 95% CI. Blue = Tier 1 (lexical), green = Tier 2 (morphological), red = Tier 3 (resistant, [R]). Negative slopes = feature declines with date; positive = rises with date.

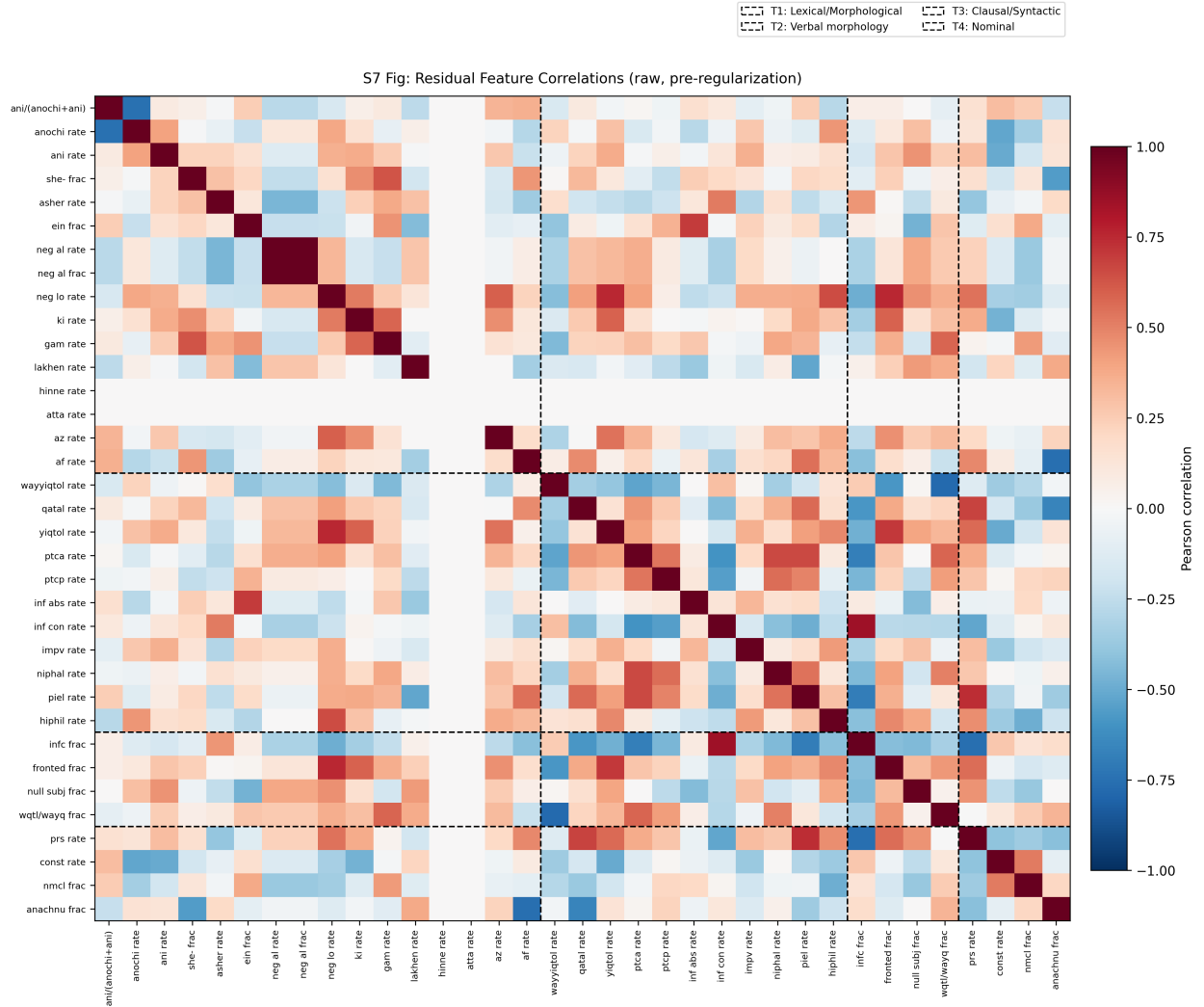


Fig 6. S7 Fig. Raw residual feature correlations (pre-regularisation). Pearson correlations among OLS residuals across the 22 training texts, computed *before* Tikhonov regularisation ($\lambda = 0.10$). Regularisation shrinks these off-diagonal entries toward zero to ensure the 35×35 covariance matrix Σ is invertible (the raw matrix is rank-deficient because $N = 22 < J = 35$). Tier boundaries are marked by dashed lines. Within-tier correlations—particularly the negative coupling between complementary forms (e.g., *wayyiqtol/qatal*, *asher/she-*)—confirm that features co-vary beyond the linear time trend, motivating the full MVN likelihood over an independent-feature approximation.

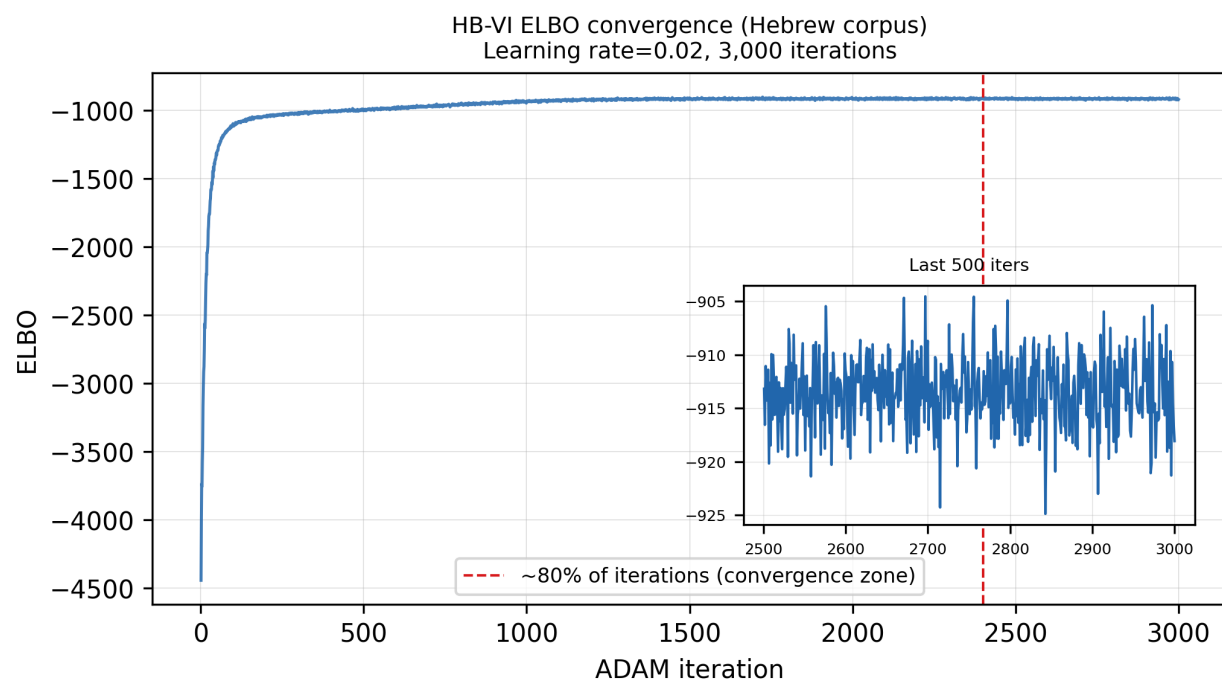


Fig 7. S9 Fig. HB-VI ELBO convergence. ELBO vs. ADAM iteration (learning rate = 0.02, 3,000 iterations). Dashed line marks the 80% convergence threshold; inset shows the final 500 iterations confirming near-flat behaviour.

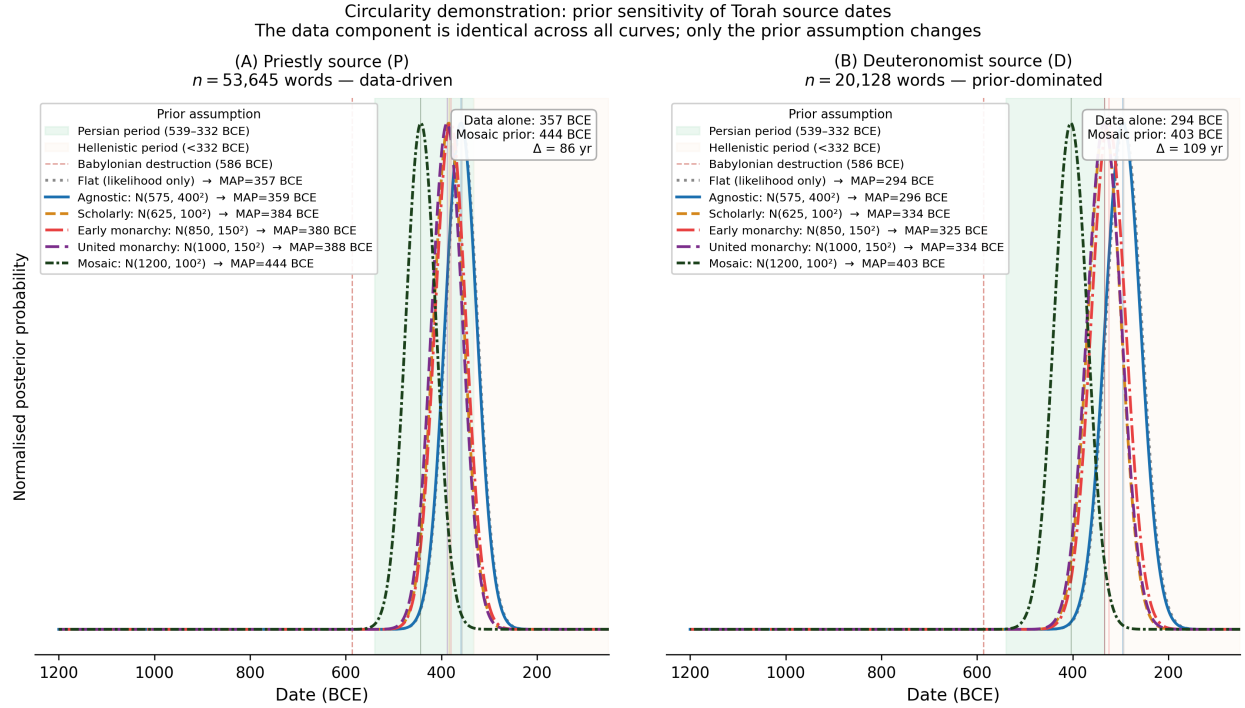


Fig 8. S14 Fig. Circularity demonstration: prior sensitivity of Torah source dates. Normalised MLE-MVN posterior curves for (A) the Priestly source ($n = 53,645$ words) and (B) the Deuteronomist source ($n = 20,128$ words), under six prior assumptions: a data-only flat prior, the agnostic prior $\mathcal{N}(575, 400^2)$, a scholarly prior $\mathcal{N}(625, 100^2)$, an early-monarchy prior $\mathcal{N}(850, 150^2)$, a united-monarchy prior $\mathcal{N}(1000, 150^2)$, and a Mosaic-authorship prior $\mathcal{N}(1200, 100^2)$. The green shading marks the Persian period (539–332 BCE); the vertical dashed red line marks the Babylonian destruction (586 BCE). Feature likelihoods are identical across all curves; only the prior assumption varies. Even the Mosaic prior yields MAP dates of 444 BCE (P) and 403 BCE (D), both firmly within the Persian period. The pre-exilic Torah date is not recoverable from the uncontaminated linguistic signal.

Table 7. S3 Table. Top dating-driving features for the three Documentary Hypothesis source composites. For each source the three features most strongly indicating late and early composition are shown with their normalized archaism scores (Eq. 6): $s = 0$ = CBH-like, $s = 1$ = LBH-like, $s < 0$ = more archaic than the SBH anchor, $s > 1$ = more LBH-like than the LBH anchor. “Genre flag” marks features whose value in this source is plausibly driven by prose genre rather than date (see text). All features are computed directly from BHSA morphological tags and can be verified without running the model. Each source heading reports both model estimates: MAP_{MVN} is the MLE-MVN posterior mode under the agnostic prior (the quantity plotted in Fig. 2), and MAP_{HB-VI} is the corresponding HB-VI estimate. The two models differ by 40–100 yr for these composites because HB-VI pools information across register groups and propagates coefficient uncertainty, which pulls prior-dominated sources toward their group mean; both models place all three sources in the Persian-to-Hellenistic period.

Source	Direction	Feature	Score	Note
<i>D source composite (Deuteronomist; $MAP_{MVN} = 292$ BCE, $MAP_{HB-VI} = 394$ BCE; prior-dominated Class P, $\Delta_{AB} = 178$ yr)</i>				
D	Late	<i>asher</i> relative particle rate	+3.31	Genre flag: high rate reflects legal-prose relative-clause density, not late date
D	Late	<i>eleh</i> “these” rate	+1.26	Consistent LBH lexical marker
D	Late	Infinitive-construct fraction	+1.25	LBH syntactic feature
D	Early	<i>anoki</i> vs. <i>ani</i> 1sg pronoun	−1.20	Classic SBH marker; archaic divine-speech convention in legal texts
D	Early	<i>sham</i> “there” rate	−1.27	Classic SBH marker
D	Early	<i>lema’an</i> “in order that” rate	−1.97	Classic SBH marker
<i>P source composite (Priestly; $MAP_{MVN} = 361$ BCE, $MAP_{HB-VI} = 404$ BCE; data-driven Class D, $\Delta_{AB} = 4$ yr)</i>				
P	Late	Niphal stem rate	+1.39	Consistent LBH morphological marker
P	Late	Abstract nominal rate	+1.32	Consistent LBH feature (high nominal density)
P	Late	<i>eleh</i> “these” rate	+1.28	Consistent LBH lexical marker
P	Early	Infinitive-construct rate	+0.15	Low inf. construct (legal/priestly prose pattern)
P	Early	<i>she-</i> relative particle rate	−0.06	Classic SBH marker; no late relative pronoun
P	Early	Construct-state noun count	−0.04	High construct ratio (SBH nominal style)
<i>JE source composite (Jahwist-Elohists; $MAP_{MVN} = 435$ BCE, $MAP_{HB-VI} = 531$ BCE; prior-dominated Class P, $\Delta_{AB} = 172$ yr)</i>				
JE	Late	Wayyiqtol fraction of narrative verbs	+1.51	Genre flag: high rate reflects narrative genre, not exclusively date
JE	Late	<i>bet</i> preposition rate	+1.42	Consistent LBH prose marker
JE	Late	<i>waw</i> conjunction rate	+1.28	Consistent LBH narrative marker
JE	Early	<i>she-</i> relative particle rate	−0.05	Classic SBH marker; no late relative pronoun
JE	Early	<i>anoki</i> vs. <i>ani</i> 1sg pronoun	−0.46	SBH marker
JE	Early	<i>sham</i> “there” rate	−0.56	Classic SBH marker

Scores are computed as $s_j = (x_j - \hat{f}_j(720 \text{ BCE})) / (\hat{f}_j(250 \text{ BCE}) - \hat{f}_j(720 \text{ BCE}))$ where $\hat{f}_j$ is the OLS-fitted feature trajectory. Scores > 1 indicate the feature value exceeds the LBH anchor level. Features marked “Genre flag” should be interpreted with caution because the training corpus contains no legal-prose texts from the Persian or Hellenistic period; see text.

Table 8. S4 Table. Robustness of source MAP dates to exclusion of genre-confounded features. MLE-MVN MAP dates for D, P, and JE under progressive feature exclusion. R1 removes *asher* frequency (genre-flagged for D; see S3 Table). R2 additionally removes wayyiqtol fraction (genre-flagged for JE). R3 removes all features with normalized archaism score $|s| > 2.5$ for any source. Feature vectors are reconstructed from normalized archaism scores (see Methods); absolute dates carry a minor approximation error, so the shift column Δ is the primary quantity of interest. R2 and R3 produce results identical to R1, confirming that *asher* is the sole high-leverage outlier. All sources remain in the Persian-to-Hellenistic period under all exclusion schemes.

Source	Baseline (BCE)	R1: drop <i>asher</i> (BCE)	Δ (yr)	R2 = R3 (BCE)	Period
D (Deuteronomist)	313	387	+74	387	Persian
P (Priestly)	347	368	+21	368	Persian
JE (Jahwist-Elohists)	442	456	+14	456	Persian/Hellenistic

Baseline dates use agnostic prior (Mode B, $\mathcal{N}(575, 400^2)$ BCE) to isolate the data signal from prior assumptions. Positive Δ indicates shift toward more recent date after feature exclusion. “Persian period” = 539–332 BCE; “Persian/Hellenistic” = 539–300 BCE. The 74 yr shift for D after excluding *asher* confirms that the dominant late signal in D is genre-confounded; the remaining Persian-period date reflects the contribution of the remaining late lexical markers (*eleh*, infinitive construct; see S3 Table).

Table 9. S5 Table. Corpus word counts for training texts and out-of-sample sources. All word counts are morphological words (BHSA) excluding coordinating conjunctions. The 19 training texts span 760–330 BCE and cover prophetic, poetic, narrative, and wisdom registers; three additional texts (Habakkuk, Haggai, Daniel) serve as cross-validation holdouts for the HB-VI model. Documentary Hypothesis source composites and sub-source divisions are excluded from both training and validation; their dates are derived solely from the model posterior. Texts below ~1,000 words produce noticeably wider posteriors due to sampling noise in feature estimates (see Fig. S10).

Text	Date (BCE)	Genre	Words	Role
<i>(A) Training corpus and cross-validation holdouts (oldest to most recent)</i>				
Amos	760 ± 15	Prophecy	2,780	Training
Hosea	725 ± 20	Prophecy	3,146	Training
Micah	720 ± 20	Prophecy	1,895	Training
Isaiah (chs. 1–39)	700 ± 15	Prophecy	13,504	Training
Zephaniah	630 ± 15	Prophecy	1,037	Training
Nahum	620 ± 20	Prophecy	746	Training
Habakkuk	605 ± 20	Prophecy	897	HB-VI holdout
Jeremiah	590 ± 15	Prophecy	14,539	Training
Lamentations	586 ± 20	Poetry	1,945	Training
Ezekiel	570 ± 15	Prophecy	26,182	Training
Deutero-Isaiah (chs. 40–55)	550 ± 20	Prophecy	5,738	Training
Haggai	520 ± 5	Prophecy	877	HB-VI holdout
Zechariah (chs. 1–8)	518 ± 5	Prophecy	2,484	Training
Trito-Isaiah (chs. 56–66)	500 ± 100	Prophecy	3,689	Training
Malachi	460 ± 20	Prophecy	1,187	Training
Jonah	400 ± 50	Narrative	985	Training
Chronicles	350 ± 30	Narrative	35,330	Training
Esther	350 ± 50	Narrative	4,621	Training
Ezra	350 ± 75	Narrative	5,268	Training
Nehemiah	350 ± 75	Narrative	7,842	Training
Ecclesiastes	330 ± 80	Wisdom	4,233	Training
Daniel	167 ± 10	Mixed	8,072	HB-VI holdout
<i>(B) Out-of-sample Documentary Hypothesis sources (excluded from all model fitting)</i>				
D source composite	—	Legal	20,128	Out-of-sample
P source composite	—	Legal	53,645	Out-of-sample
JE source composite	—	Narrative	40,867	Out-of-sample
D Code (Deut 12–26)	—	Legal	7,353	Out-of-sample
D Frame (Deut 1–11, 27–34)	—	Narrative	11,993	Out-of-sample
D Song (Deut 32–33)	—	Other	782	Out-of-sample
Lev Holiness Code (17–26)	—	Legal	5,958	Out-of-sample
Lev Priestly (1–16)	—	Legal	10,496	Out-of-sample

HB-VI holdouts (Habakkuk, Haggai, Daniel) are withheld from *all* model fitting; their MAP dates assess cross-validation accuracy only (Fig. 9). The training/holdout distinction applies to HB-VI; all 22 texts in panel (A) appear in MLE-MVN training. D Song (782 words) should be treated with caution given its small size and poetic genre. CBH = Classical Biblical Hebrew; LBH = Late Biblical Hebrew; Transitional = texts with mixed register features.

Table 10. S6 Table. Prior-sweep MAP dates for the three Torah source composites. MLE-MVN MAP dates (BCE) for the Priestly (P), Deuteronomist (D), and Yahwist-Elohlist (JE) source composites under six prior scenarios ranging from an uninformative flat (data-only) prior to a Mosaic-authorship prior $\mathcal{N}(1200, 100^2)$. Feature vectors are held fixed across all rows (reconstructed from normalized archaism scores; see Methods and Fig. S14); only the prior assumption changes. The Babylonian destruction (586 BCE), the Cyrus decree (539 BCE), and the onset of the Hellenistic period (332 BCE) are shown for reference. None of the six scenarios recovers a pre-exilic MAP date for any source; the data signal is strong enough to anchor all three sources in the Persian period even under the most extreme traditional prior.

Prior scenario	μ_0 (BCE)	σ_0 (yr)	P MAP	D MAP	JE MAP	Period
Flat (data only)	—	—	357	294	434	Persian
Agnostic $\mathcal{N}(575, 400^2)$	575	400	359	296	434	Persian
Scholarly $\mathcal{N}(625, 100^2)$	625	100	384	334	453	Persian
Early monarchy $\mathcal{N}(850, 150^2)$	850	150	380	325	453	Persian
United monarchy $\mathcal{N}(1000, 150^2)$	1000	150	388	334	461	Persian
Mosaic $\mathcal{N}(1200, 100^2)$	1200	100	444	403	511	Persian

Feature vectors reconstructed from normalized archaism scores (archaism_by_feature.csv) via OLS back-transform; see Methods. Word-count scaling $\text{clip}(n/5000, 1, 5)$ applied (P $n = 53,645$; D $n = 20,128$; JE $n = 40,867$ words). The P–Mosaic shift of 87 yr across the full prior range confirms the data-driven classification from the prior sensitivity analysis ($|\Delta_{AB}| = 4$ yr; Table 3).